Unit 7. Unsupervised Learning

Business Data Analytics: Introductory Machine Learning

Without labels

Units 2–6 fit functions $f(x) \approx y$ from labeled pairs (x_i, y_i). Unit 7 starts from $X \in \mathbb{R}^{n\times p}$ alone, no y, and asks two questions: which low-dimensional view of the rows is most informative (PCA, t-SNE, UMAP), and which rows belong together (K-means, hierarchical clustering). Without a target, every method is judged by what its output reveals, not by held-out accuracy. The pitfalls section is correspondingly the largest in the course.

1 Preliminaries

1.1 Spectral theorem and Courant–Fischer min-max

Unit 2 §2.1.1 proved the spectral theorem and its Rayleigh-quotient corollary. PCA reuses both; the Eckart–Young proof in §1.3 additionally needs the subspace generalization due to Courant and Fischer.

Definition 7.1 (Eigenvalue, eigenvector, orthogonal matrix)**.** For symmetric $A \in \mathbb{R}^{p\times p}$, a scalar $\lambda \in \mathbb{R}$ is an **eigenvalue** with **eigenvector** $v \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ when $Av = \lambda v$. (Reality of every eigenvalue of a real symmetric A is part of the spectral theorem stated below; for non-symmetric real A, eigenvalues and eigenvectors may be complex.) A matrix $Q \in \mathbb{R}^{p\times p}$ is **orthogonal** when $Q^\top Q = I_p$; equivalently, its columns are orthonormal. For square Q, orthonormal columns force orthonormal rows (Unit 2 Lemma 2.4), so $Q^{-1} = Q^\top$.

Theorem 7.2 (Spectral theorem; Unit 2 §2.1.1, Theorem 2.6)**.** Every symmetric $A \in \mathbb{R}^{p\times p}$ admits an orthogonal diagonalization

$$A = Q\Lambda Q^\top, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \quad \lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_p,$$

with real eigenvalues and orthonormal eigenvectors $q_1, \ldots, q_p$ in the columns of Q.

The Unit 2 proof constructs each q_j as the argmax of $x^\top Ax$ over $\{x \in W_j : \|x\| = 1\}$ with $W_j = \operatorname{span}(q_1, \ldots, q_{j-1})^\perp$, starting from $W_1 = \mathbb{R}^p$. The Rayleigh-quotient corollary (Unit 2 Theorem 2.7) reads this off: λ_k is the maximum of $x^\top Ax$ over unit vectors orthogonal to $q_1, \ldots, q_{k-1}$, attained at q_k. The next theorem replaces the orthogonal-complement constraint with a min-max over arbitrary k-dimensional subspaces, needed for the Eckart–Young proof of §1.3.

Theorem 7.3 (Courant–Fischer min-max)**.** For symmetric $A \in \mathbb{R}^{p\times p}$ with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_p$,

$$\lambda_k = \max_{\substack{T\subseteq\mathbb{R}^p \\ \dim T=k}} \min_{\substack{x\in T \\ \|x\|=1}} x^\top Ax = \min_{\substack{T\subseteq\mathbb{R}^p \\ \dim T=p-k+1}} \max_{\substack{x\in T \\ \|x\|=1}} x^\top Ax.$$

Proof of Theorem 7.3

Write the spectral decomposition $A = Q\Lambda Q^\top$ as in Theorem 7.2. We prove the first equality; the second follows by applying the first to $-A$ (whose eigenvalues are $-\lambda_p \geqslant \cdots \geqslant -\lambda_1$) and reading off $-\lambda_{p-k+1}$.

Lower bound. Choose $T = \operatorname{span}(q_1, \ldots, q_k)$, dimension k. For unit $x \in T$, $x = \sum_{j=1}^k c_j q_j$ with $\sum_j c_j^2 = 1$, so

$$x^\top Ax = \sum_{j=1}^{k} c_j^2\lambda_j \geqslant \lambda_k \sum_{j=1}^{k} c_j^2 = \lambda_k.$$

Hence $\min_{x\in T,\, \|x\|=1} x^\top Ax \geqslant \lambda_k$, and the maximum over k-subspaces is at least λ_k.

Upper bound. Let $T \subseteq \mathbb{R}^p$ be any k-dimensional subspace. Set $L = \operatorname{span}(q_k, q_{k+1}, \dots, q_p)$, dimension $p - k + 1$ (a different letter from the sample covariance S that will appear in §1.4 onward). By the dimension formula

$$\dim(T \cap L) \;=\; \dim T + \dim L - \dim(T + L) \;\geqslant\; k + (p - k + 1) - p \;=\; 1,$$

so $T \cap L$ contains a unit vector x. Expand $x = \sum_{j \geqslant k} c_j q_j$ with $\sum_{j \geqslant k} c_j^2 = 1$:

$$x^\top A x \;=\; \sum_{j \geqslant k} c_j^2 \lambda_j \;\leqslant\; \lambda_k.$$

Hence $\min_{x \in T,\, \|x\|=1} x^\top A x \leqslant \lambda_k$. Since T was arbitrary, the maximum over k-subspaces is at most λ_k.

The two bounds give equality. ∎

1.2 Singular value decomposition

The spectral theorem applies to symmetric matrices. PCA's data matrix $X \in \mathbb{R}^{n\times p}$ is rectangular. The singular value decomposition extends the spectral picture to rectangular matrices using two orthogonal frames, one in $\mathbb{R}^n$ and one in $\mathbb{R}^p$.

Definition 7.4 (Singular value decomposition)**.** A **(full) singular value decomposition** of $X \in \mathbb{R}^{n\times p}$ is a factorization

$$X = U\Sigma V^\top,$$

where $U \in \mathbb{R}^{n\times n}$ and $V \in \mathbb{R}^{p\times p}$ are orthogonal and $\Sigma \in \mathbb{R}^{n\times p}$ is diagonal with nonnegative entries $\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_{\min(n,p)} \geqslant 0$ on the leading diagonal and zeros elsewhere. The σ_j are the **singular values**; the columns of U and V are the **left** and **right singular vectors**.

Theorem 7.5 (Existence of the SVD)**.** Every $X \in \mathbb{R}^{n\times p}$ admits a singular value decomposition. The singular values are uniquely determined; they coincide with $\sqrt{\lambda_j(X^\top X)}$ where $\lambda_j(\cdot)$ denotes the j-th largest eigenvalue.

Proof of Theorem 7.5

The Gram matrix $X^\top X \in \mathbb{R}^{p\times p}$ is symmetric. For any $v \in \mathbb{R}^p$, $v^\top X^\top X v = \|Xv\|^2 \geqslant 0$, so $X^\top X \succcurlyeq 0$ (Unit 1 §1.9.6). Apply Theorem 7.2: there exist orthonormal $v_1, \dots, v_p \in \mathbb{R}^p$ and $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$ with $X^\top X v_j = \lambda_j v_j$. Set $\sigma_j = \sqrt{\lambda_j}$ and let r be the largest index with $\sigma_r > 0$, so $\sigma_{r+1} = \cdots = \sigma_p = 0$.

Construct U. For $j = 1, \dots, r$, define $u_j = Xv_j/\sigma_j \in \mathbb{R}^n$. These are orthonormal: for $1 \leqslant i, j \leqslant r$,

$$u_i^\top u_j = \frac{(Xv_i)^\top (Xv_j)}{\sigma_i \sigma_j} = \frac{v_i^\top X^\top X v_j}{\sigma_i \sigma_j} = \frac{\sigma_j^2 v_i^\top v_j}{\sigma_i \sigma_j} = \delta_{ij}.$$

Extend $\{u_1, \dots, u_r\}$ to an orthonormal basis $\{u_1, \dots, u_n\}$ of $\mathbb{R}^n$ (Gram–Schmidt on any complementary basis).

Verify the factorization. Set $U = [u_1 \ \cdots \ u_n]$, $V = [v_1 \ \cdots \ v_p]$, and Σ the $n \times p$ matrix with $\Sigma_{jj} = \sigma_j$ for $j = 1, \dots, \min(n, p)$ and 0 elsewhere. For $j \leqslant r$, $Xv_j = \sigma_j u_j$ by construction. For $j > r$, $X^\top X v_j = \mathbf{0}$ implies $\|Xv_j\|^2 = v_j^\top X^\top X v_j = 0$, so $Xv_j = \mathbf{0} = \sigma_j u_j$. The columnwise identity $XV = U\Sigma$ then holds (with the appropriate column-index alignment in Σ), and rightmultiplying by $V^\top$ gives $X = U\Sigma V^\top$.

Uniqueness of σ_j. Any SVD $X = U\Sigma V^\top$ gives $X^\top X = V\Sigma^\top\Sigma V^\top = V\operatorname{diag}(\sigma_1^2, \ldots, \sigma_p^2)V^\top$, so each σ_j^2 equals the j-th eigenvalue of $X^\top X$ and is determined by X alone. The convention $\sigma_j \geqslant 0$ in Definition 7.4 then selects $\sigma_j = +\sqrt{\sigma_j^2}$ from the two square roots, so σ_j is determined by X. ■

Reduced and thin SVD

When $n > p$, the last $n-p$ columns of U contribute nothing: the corresponding rows of Σ are zero. Dropping them gives the **reduced** SVD $X = U_p\Sigma_p V^\top$ with $U_p \in \mathbb{R}^{n\times p}$, $\Sigma_p = \operatorname{diag}(\sigma_1, \ldots, \sigma_p)$, $V \in \mathbb{R}^{p\times p}$. PCA implementations (`prcomp` in R) compute the reduced form. Whenever the distinction is irrelevant, the text writes $X = U\Sigma V^\top$ without specifying full or reduced.

Lemma 7.6 (Frobenius norm in singular values)**.** For $X \in \mathbb{R}^{n\times p}$ with singular values $\sigma_1 \geqslant \cdots \geqslant$

$\sigma_{\min(n,p)} \geqslant 0$,

$$\|X\|_F^2 := \sum_{i,j} X_{ij}^2 = \operatorname{tr}(X^\top X) = \sum_j \sigma_j^2.$$

Proof. $\sum_{i,j} X_{ij}^2 = \sum_j \|X_{\cdot,j}\|^2 = \sum_j (X^\top X)_{jj} = \operatorname{tr}(X^\top X)$. The trace is invariant under similarity, so $\operatorname{tr}(X^\top X) = \operatorname{tr}(V \operatorname{diag}(\sigma_j^2) V^\top) = \operatorname{tr}(\operatorname{diag}(\sigma_j^2)) = \sum_j \sigma_j^2$. □

1.3 Eckart–Young low-rank approximation

Among all matrices of rank at most k, the one closest to X in Frobenius distance is the truncated SVD that retains the k largest singular values.

Definition 7.7 (Truncated SVD)**.** For $X = U\Sigma V^\top$ as in Definition 7.4 and $k \leqslant \min(n,p)$, the **rank-k truncated SVD** of X is

$$X_k := \sum_{j=1}^{k} \sigma_j u_j v_j^\top = U_k \Sigma_k V_k^\top,$$

where $U_k = [u_1 \ \cdots \ u_k]$, $V_k = [v_1 \ \cdots \ v_k]$, $\Sigma_k = \operatorname{diag}(\sigma_1, \ldots, \sigma_k)$.

Theorem 7.8 (Eckart–Young, Frobenius norm)**.** For $X \in \mathbb{R}^{n\times p}$ with singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_{\min(n,p)} \geqslant 0$ and any $k \leqslant \min(n,p)$,

$$\min_{\substack{B\in\mathbb{R}^{n\times p}\\ \operatorname{rank} B\leqslant k}} \|X-B\|_F^2 \;=\; \sum_{j>k} \sigma_j^2,$$

attained at $B^\star = X_k$, the rank-k truncated SVD of X.

Proof of Theorem 7.8

Achievability. Direct expansion gives

$$X - X_k \;=\; \sum_{j>k} \sigma_j u_j v_j^\top, \qquad \|X-X_k\|_F^2 \;=\; \sum_{j>k} \sigma_j^2,$$

the second equality by Lemma 7.6 applied to $X - X_k$ (whose singular values are $\sigma_{k+1}, \ldots, \sigma_{\min(n,p)}, 0, \ldots, 0$).

Lower bound: singular-value interlacing. Let $B \in \mathbb{R}^{n\times p}$ with $\operatorname{rank} B \leqslant k$. We claim $\sigma_i(X-B) \geqslant \sigma_{i+k}(X)$ for every i such that $i+k \leqslant \min(n,p)$ (with the convention $\sigma_j(\cdot) = 0$

when j exceeds the matrix size).
Set $M = X - B$. Since $M^\top M$ is symmetric PSD, by Theorem 7.3

$$\sigma_i(M)^2 \;=\; \lambda_i(M^\top M) \;=\; \max_{\substack{T\subseteq\mathbb{R}^p\\ \dim T=i}} \; \min_{\substack{x\in T\\ \|x\|=1}} \|Mx\|^2.$$

We exhibit a specific i-dimensional T on which $\|Mx\|^2 \geqslant \sigma_{i+k}(X)^2$ for every unit $x \in T$.
Let $\mathcal{V}_{i+k} = \operatorname{span}(v_1, \dots, v_{i+k})$ (a subspace, distinct from any V-matrix), where v_j are right singular vectors of X, $\dim \mathcal{V}_{i+k} = i + k$. The kernel of B has dimension $\geqslant p - k$. By the dimension formula,

$$\dim\big(\mathcal{V}_{i+k} \cap \ker B\big) \;\geqslant\; (i+k) + (p-k) - p \;=\; i.$$

Pick any i-dimensional subspace $T \subseteq \mathcal{V}_{i+k} \cap \ker B$. For unit $x \in T$, $Bx = \mathbf{0}$, so

$$\|Mx\|^2 \;=\; \|Xx - Bx\|^2 \;=\; \|Xx\|^2.$$

Expand $x = \sum_{j=1}^{i+k} c_j v_j$ with $\sum c_j^2 = 1$. Using $Xv_j = \sigma_j u_j$ and orthonormality of u_j,

$$\|Xx\|^2 = \sum_{j=1}^{i+k} c_j^2 \sigma_j^2 \geqslant \sigma_{i+k}^2 \sum_{j=1}^{i+k} c_j^2 = \sigma_{i+k}^2.$$

Hence $\min_{x \in T,\, \|x\|=1} \|Mx\|^2 \geqslant \sigma_{i+k}^2$, so $\sigma_i(M)^2 \geqslant \sigma_{i+k}^2$, proving the claim.

Sum over i. By Lemma 7.6,

$$\|X - B\|_F^2 = \sum_{i \geqslant 1} \sigma_i(X-B)^2 \geqslant \sum_{i \geqslant 1} \sigma_{i+k}(X)^2 = \sum_{j > k} \sigma_j(X)^2,$$

matching the achieved value. ■

Beyond Frobenius

Eckart–Young also holds in the spectral norm $\|\cdot\|_2$ and in every unitarily invariant norm; this generalization is Mirsky's theorem (Mirsky, 1960). PCA only uses the Frobenius statement, which is what Theorem 7.8 delivers.

1.4 Sample mean, centering matrix, sample covariance

PCA is computed on the centered data matrix; centering is a linear operation worth recording in matrix form.

Definition 7.9 (Sample mean). For a data matrix $X \in \mathbb{R}^{n\times p}$ with rows $x_1^\top, \dots, x_n^\top$, the **sample mean** is

$$\bar{x} \;:=\; \frac{1}{n}\sum_{i=1}^{n} x_i \;=\; \frac{1}{n}X^\top \mathbf{1}_n \;\in\; \mathbb{R}^p,$$

where $\mathbf{1}_n \in \mathbb{R}^n$ is the all-ones vector.

Definition 7.10 (Centering matrix). The **centering matrix** of order n is

$$\mathbf{M}_0 \;:=\; I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top \;\in\; \mathbb{R}^{n\times n}.$$

Lemma 7.11 (Properties of $\mathbf{M}_0$).

(i) $\mathbf{M}_0 = \mathbf{M}_0^\top$ (symmetric).

(ii) $\mathbf{M}_0^2 = \mathbf{M}_0$ (idempotent).

(iii) $\mathbf{M}_0 \mathbf{1}_n = \mathbf{0}$ and $\mathbf{1}_n^\top \mathbf{M}_0 = \mathbf{0}^\top$.

(iv) Eigenvalues of $\mathbf{M}_0$ are 0 (multiplicity 1, eigenvector $\mathbf{1}_n/\sqrt{n}$) and 1 (multiplicity $n-1$); $\operatorname{rank} \mathbf{M}_0 = n - 1$.

(v) For any $X \in \mathbb{R}^{n\times p}$, the rows of $\mathbf{M}_0 X$ are $x_i - \bar{x}$:

$$\mathbf{M}_0 X \;=\; X - \mathbf{1}_n \bar{x}^\top.$$

Proof. (i) Both I_n and $\mathbf{1}_n\mathbf{1}_n^\top$ are symmetric. (ii) Compute

$$\mathbf{M}_0^2 \;=\; \left(I_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\right)^2 \;=\; I_n - \tfrac{2}{n}\mathbf{1}_n\mathbf{1}_n^\top + \tfrac{1}{n^2}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{1}_n\mathbf{1}_n^\top \;=\; I_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top \;=\; \mathbf{M}_0,$$

using $\mathbf{1}_n^\top\mathbf{1}_n = n$. (iii) $\mathbf{M}_0\mathbf{1}_n = \mathbf{1}_n - \frac{1}{n}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n) = \mathbf{1}_n - \mathbf{1}_n = \mathbf{0}$. (iv) Idempotents have eigenvalues in $\{0,1\}$; $\mathbf{1}_n/\sqrt{n}$ is the unique-up-to-sign eigenvector with eigenvalue 0 by (iii); the orthogonal complement (dimension $n-1$) is the eigenspace for 1. (v) $\mathbf{M}_0 X = X - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top X$, and $\mathbf{1}_n^\top X/n = \bar{x}^\top$

by Definition 7.9, so $\mathbf{M}_0 X = X - \mathbf{1}_n \bar{x}^\top$, whose row i is $x_i - \bar{x}$. □

Definition 7.12 (Sample covariance matrix)**.** The **sample covariance matrix** of $X \in \mathbb{R}^{n\times p}$ is

$$S := \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^\top = \frac{1}{n-1}(\mathbf{M}_0 X)^\top(\mathbf{M}_0 X) = \frac{1}{n-1}X^\top \mathbf{M}_0 X,$$

the last equality using $\mathbf{M}_0^\top \mathbf{M}_0 = \mathbf{M}_0^2 = \mathbf{M}_0$ (Lemma 7.11). S is symmetric and PSD.

n versus $n-1$

The denominator $n-1$ makes S unbiased for $\operatorname{cov}(x_1)$ when the x_i are i.i.d. from a distribution with finite second moments. PCA derivations are insensitive to the choice: scaling S by a positive constant rescales eigenvalues but does not move eigenvectors, so the principal directions are identical for the n and $n-1$ versions. The course uses $n-1$ throughout to match `var`, `cov`, and `prcomp` in R.

1.5 KL divergence and fuzzy cross-entropy

Unit 3 §3.1.1 introduced the Kullback–Leibler divergence and cross-entropy for probability mass functions on a finite outcome set; logistic regression was the first application. t-SNE reuses both verbatim. UMAP needs a generalization in which each pair of data points carries its own Bernoulli pair of probabilities, applied to fuzzy-set membership values rather than probability mass functions.

Definition 7.13 (KL divergence; Unit 3 Definition 3.5)**.** For pmfs p, q on a finite set $\mathcal{Y}$ with $q(y) > 0$ wherever $p(y) > 0$,

$$D_{\mathrm{KL}}(p \,\|\, q) \;:=\; \sum_{y \in \mathcal{Y}} p(y) \log \frac{p(y)}{q(y)}, \qquad 0 \log 0 := 0.$$

Lemma 7.14 (Gibbs's inequality; Unit 3 Lemma 3.6)**.** $D_{\mathrm{KL}}(p \,\|\, q) \geqslant 0$, with equality iff $p = q$.

Proof. Let $\operatorname{supp}(p) = \{y : p(y) > 0\}$. By Jensen's inequality (Unit 1 §1.9.9) for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{y \in \operatorname{supp}(p)} p(y) \log \frac{q(y)}{p(y)} \leqslant \log \sum_{y \in \operatorname{supp}(p)} q(y) \leqslant \log 1 = 0.$$

Equality in Jensen's requires $q(y)/p(y)$ constant on $\operatorname{supp}(p)$; combined with both sums equaling 1, this forces $p = q$. □

Definition 7.15 (Cross-entropy; Unit 3 Definition 3.7)**.** For pmfs p, q on $\mathcal{Y}$, the **cross-entropy** from p to q is $H(p, q) := -\sum_y p(y) \log q(y)$, and the **Shannon entropy** of p is $H(p) := H(p, p)$. Algebraic manipulation gives $H(p, q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$. With p fixed, $\arg\min_q H(p, q) = \arg\min_q D_{\mathrm{KL}}(p \,\|\, q)$, since $H(p)$ does not depend on q.

UMAP needs a finer object than a probability distribution. The pairwise membership matrix carries one number $\mu_{ij} \in [0, 1]$ for each pair of points; these numbers do not sum to 1 and are not jointly normalized. Each μ_{ij} is treated as the success probability of an independent Bernoulli trial; the natural divergence between two such matrices sums per-pair Bernoulli KL divergences.

Definition 7.16 (Bernoulli KL)**.** For $\mu, \nu \in (0,1)$, the **Bernoulli KL divergence** between $\mathrm{Ber}(\mu)$ and $\mathrm{Ber}(\nu)$ is

$$d_{\mathrm{B}}(\mu,\nu) \;:=\; \mu \log\frac{\mu}{\nu} \;+\; (1-\mu)\log\frac{1-\mu}{1-\nu}.$$

This is Definition 7.13 applied to the two-point distributions $p = (\mu, 1-\mu)$ and $q = (\nu, 1-\nu)$ on $\{1, 0\}$.

Lemma 7.17 (Properties of d_{B})**.**

(i) $d_{\mathrm{B}}(\mu,\nu) \geqslant 0$, with equality iff $\mu = \nu$.

(ii) $d_{\mathrm{B}}(\mu,\nu) \neq d_{\mathrm{B}}(\nu,\mu)$ in general.

(iii) For each fixed $\mu \in (0,1)$, the map $\nu \mapsto d_{\mathrm{B}}(\mu,\nu)$ is strictly convex on $(0,1)$.

Proof. (i) is Lemma 7.14 with $|\mathcal{Y}| = 2$. (ii) by example: $d_{\mathrm{B}}(0.1, 0.5) = 0.1\log 0.2 + 0.9\log 1.8 \approx 0.368$ while $d_{\mathrm{B}}(0.5, 0.1) = 0.5\log 5 + 0.5\log(5/9) \approx 0.510$. (iii) Differentiate twice in ν:

$$\frac{\partial d_{\mathrm{B}}}{\partial \nu} \;=\; -\frac{\mu}{\nu} + \frac{1-\mu}{1-\nu}, \qquad \frac{\partial^2 d_{\mathrm{B}}}{\partial \nu^2} \;=\; \frac{\mu}{\nu^2} + \frac{1-\mu}{(1-\nu)^2} \;>\; 0$$

on $(0,1)$. $\square$

Definition 7.18 (Fuzzy cross-entropy)**.** Let A be a finite set and $\mu, \nu : A \to (0,1)$. The **fuzzy cross-entropy** of μ relative to ν is

$$C(\mu,\nu) \;:=\; \sum_{a\in A} d_{\mathrm{B}}(\mu(a),\nu(a)) \;=\; \sum_{a\in A}\left[\mu(a)\log\frac{\mu(a)}{\nu(a)} \;+\; (1-\mu(a))\log\frac{1-\mu(a)}{1-\nu(a)}\right].$$

$C(\mu,\nu) \geqslant 0$ by Lemma 7.17(i), with equality iff $\mu = \nu$ pointwise.

When A is the set of unordered pairs $\{i,j\}$ in $\{1,\ldots,n\}$ and μ,ν are interpreted as edge-strength functions on the complete graph K_n, the sum $C(\mu,\nu)$ is the UMAP optimization target derived in §7.5.7.

Why fuzzy cross-entropy and not KL

A pairwise membership μ_{ij} is not the probability of "selecting edge $\{i,j\}$"; it is the probability that the edge "exists". The natural model is one Bernoulli per edge. The KL divergence between two product Bernoulli laws over independent edges decomposes as a sum of per-edge d_{B}, which is exactly $C(\mu, \nu)$. Independence across pairs is a UMAP modeling assumption, not a derived fact.

1.6 Simplicial complexes

UMAP describes the data as a discrete shape whose vertices are the data points and whose higher faces encode local neighborhood relations.

Definition 7.19 (Abstract simplicial complex)**.** Let V be a finite set. An **abstract simplicial complex** on V is a collection $K \subseteq 2^V \setminus \{\varnothing\}$ closed under taking nonempty subsets:

$$\sigma \in K \text{ and } \varnothing \neq \tau \subseteq \sigma \implies \tau \in K.$$

Each $\sigma \in K$ is a **simplex** (or **face**); $|\sigma| - 1$ is its **dimension**. The dimension of K is $\max_{\sigma \in K}(|\sigma| - 1)$.

A 0-simplex is a vertex, a 1-simplex an edge, a 2-simplex a (filled) triangle, a 3-simplex a (filled) tetrahedron.

Example 7.1. Take $V = \{1,2,3\}$. The collection

$$K = \{\{1\},\{2\},\{3\},\{1,2\},\{2,3\},\{1,3\},\{1,2,3\}\}$$

is an abstract simplicial complex of dimension 2: the closed triangle. Removing $\{1,2,3\}$ yields the boundary, dimension 1.

The geometric counterpart fills each simplex with a Euclidean simplex.

Definition 7.20 (Affine independence; geometric simplex)**.** Points $w_0, \ldots, w_n \in \mathbb{R}^N$ are **affinely independent** when the differences $w_1 - w_0, \ldots, w_n - w_0$ are linearly independent in $\mathbb{R}^N$. The **geometric n-simplex** they span is the convex hull

$$[w_0, \ldots, w_n] := \Big\{ \textstyle\sum_{i=0}^n t_i w_i : t_i \geqslant 0,\ \sum_{i=0}^n t_i = 1 \Big\}.$$

The **standard n-simplex** is

$$\Delta^n := \big\{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_i \geqslant 0,\ \textstyle\sum_i t_i = 1\big\}.$$

The Δ^n inherit two families of structural maps. The **coface** maps $d^i : \Delta^{n-1} \to \Delta^n$ include Δ^{n-1} as the boundary face opposite the i-th vertex; the **codegeneracy** maps $s^i : \Delta^{n+1} \to \Delta^n$ collapse the edge between the i-th and $(i+1)$-th vertices:

$$\begin{aligned} d^i(t_0, \dots, t_{n-1}) &= (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}), \\ s^i(t_0, \dots, t_{n+1}) &= (t_0, \dots, t_{i-1}, t_i + t_{i+1}, t_{i+2}, \dots, t_{n+1}). \end{aligned}$$

These satisfy a fixed list of identities, the **simplicial identities**, recorded in §1.7 once the categorical language is in place.

Definition 7.21 (Geometric realization)**.** Let K be an abstract simplicial complex on $V = \{v_0, \dots, v_m\}$. Choose any affinely independent $w_0, \dots, w_m$ in some $\mathbb{R}^N$; e.g. take $w_i = e_{i+1}$, the $(i+1)$-th standard basis vector in $\mathbb{R}^{m+1}$. The **geometric realization** of K is

$$|K| := \bigcup_{\sigma \in K} \left[\, w_i : v_i \in \sigma \,\right] \subseteq \mathbb{R}^N,$$

the union of geometric simplices over all faces of K. The combinatorial structure of K determines $|K|$ up to homeomorphism.

UMAP works with a more flexible variant in which each simplex carries a numerical weight in $(0,1]$, the strength of its membership.

1.7 Categories, functors, and adjunctions

UMAP's mathematical foundation (Spivak, 2009; McInnes et al., 2018) describes a metric space and a fuzzy simplicial set as objects of two categories related by an adjoint pair of functors.

Definition 7.22 (Category)**.** A **category** $\mathcal{C}$ consists of:

(i) a class $\mathrm{Ob}(\mathcal{C})$ of **objects**;

(ii) for each ordered pair (A, B) of objects, a set $\hom_{\mathcal{C}}(A, B)$ of **morphisms** from A to B (also written $f : A \to B$);

(iii) for each triple (A, B, C), a **composition** map

$$\circ : \hom_{\mathcal{C}}(B, C) \times \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{C}}(A, C), \qquad (g, f) \mapsto g \circ f;$$

(iv) for each A, an **identity** morphism $1_A \in \hom_{\mathcal{C}}(A, A)$;

satisfying **associativity** ($h \circ (g \circ f) = (h \circ g) \circ f$ whenever defined) and the **unit laws** ($f \circ 1_A = f$ and $1_B \circ f = f$ for $f : A \to B$).

Example 7.2 (Standard categories).

(i) **Set**: objects are sets, morphisms are functions; composition is function composition; identities are identity functions.

(ii) $\mathbf{Vec}_{\mathbb{R}}$: objects are real vector spaces, morphisms are linear maps.

(iii) **Top**: objects are topological spaces, morphisms are continuous functions.

(iv) Any partially ordered set $(P, \leqslant)$ becomes a category $\mathcal{P}$ with $\mathrm{Ob}(\mathcal{P}) = P$ and $|\hom_{\mathcal{P}}(a, b)| = 1$ when $a \leqslant b$, 0 otherwise. Associativity is transitivity; the identity is reflexivity.

(v) **Met**: objects are metric spaces, morphisms are short maps $(d(f(x), f(y)) \leqslant d(x, y))$.

Definition 7.23 (Opposite category). The **opposite category** $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, with $\hom_{\mathcal{C}^{\mathrm{op}}}(A, B) := \hom_{\mathcal{C}}(B, A)$ and composition $g \circ_{op} f := f \circ_{\mathcal{C}} g$. Identities are identical to those of $\mathcal{C}$. The construction is involutive: $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Definition 7.24 (The simplex category). The **simplex category** Δ has:

- objects $[n] := \{0, 1, \ldots, n\}$ for $n \geqslant 0$, treated as totally ordered sets;

- morphisms $[m] \to [n]$ the order-preserving maps $f : [m] \to [n]$ (so $i \leqslant j \Rightarrow f(i) \leqslant f(j)$);
- composition is composition of maps; identities are identity maps.

The geometric maps d^i, s^i from §1.6 correspond to morphisms in Δ:

$$d^i : [n-1] \to [n], \qquad d^i(j) = \begin{cases} j & j < i \\ j+1 & j \geqslant i \end{cases} \qquad \text{(skip } i\text{)},$$

$$s^i : [n+1] \to [n], \qquad s^i(j) = \begin{cases} j & j \leqslant i \\ j-1 & j > i \end{cases} \qquad \text{(double } i\text{)}.$$

Every morphism in Δ factors as a composition of these face inclusions and degeneracy surjections; the relations among them are the simplicial identities (Mac Lane, 1971, §VII.5).

Definition 7.25 (Functor)**.** A **functor** $F : \mathcal{C} \to \mathcal{D}$ assigns:

- to each $A \in \mathrm{Ob}(\mathcal{C})$ an object $FA \in \mathrm{Ob}(\mathcal{D})$;
- to each morphism $f : A \to B$ in $\mathcal{C}$ a morphism $Ff : FA \to FB$ in $\mathcal{D}$;

preserving composition ($F(g \circ f) = Fg \circ Ff$) and identities ($F1_A = 1_{FA}$). A functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is a **contravariant functor**; explicitly, it sends $f : A \to B$ in $\mathcal{C}$ to $Ff : FB \to FA$ in $\mathcal{D}$ and reverses the composition order.

Example 7.3 (Standard functors)**.**

(i) Identity: $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$.

(ii) Forgetful: $U : \mathbf{Vec}_{\mathbb{R}} \to \mathbf{Set}$ sends a vector space to its underlying set, a linear map to its underlying function.

(iii) Free: $F : \mathbf{Set} \to \mathbf{Vec}_{\mathbb{R}}$ sends a set A to the free vector space $\mathbb{R}^A$ and a function $f : A \to B$ to its unique linear extension Ff.

(iv) Hom-functor: For fixed $A \in \mathcal{C}$, $\hom_{\mathcal{C}}(A, -) : \mathcal{C} \to \mathbf{Set}$ sends $B \mapsto \hom_{\mathcal{C}}(A, B)$ and $f : B \to B'$ to post-composition $f \circ - : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{C}}(A, B')$.

(v) Presheaf: a functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is a **presheaf on** $\mathcal{C}$; the category of presheaves is $\mathbf{PSh}(\mathcal{C}) := \mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}$. **Simplicial sets** are presheaves on Δ: $\mathbf{sSet} := \mathbf{Set}^{\Delta^{\mathrm{op}}}$.

A simplicial set $X \in \mathbf{sSet}$ assigns to each $[n]$ a set $X_n := X([n])$ of n**-simplices**, and to each order-preserving $f : [m] \to [n]$ a map $X(f) : X_n \to X_m$ (note the reversal: X is contravariant on

Δ). The face and degeneracy operators $d_i := X(d^i) : X_n \to X_{n-1}$ and $s_i := X(s^i) : X_n \to X_{n+1}$ are the data the abstract simplicial complex of §1.6 carries when reformulated functorially.

Definition 7.26 (Natural transformation)**.** For functors $F, G : \mathcal{C} \to \mathcal{D}$, a **natural transformation** $\eta : F \Rightarrow G$ is a family of morphisms $\eta_A : FA \to GA$ in $\mathcal{D}$, indexed by $A \in \mathrm{Ob}(\mathcal{C})$, such that for every $f : A \to B$ in $\mathcal{C}$ the square

$$\begin{array}{ccc} FA & \xrightarrow{\eta_A} & GA \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ FB & \xrightarrow{\eta_B} & GB \end{array}$$

commutes: $Gf \circ \eta_A = \eta_B \circ Ff$. The morphisms η_A are the **components** of η.

The composition of natural transformations and the identity 1_F (with $(1_F)_A = 1_{FA}$) make functors $\mathcal{C} \to \mathcal{D}$ into a category $[\mathcal{C}, \mathcal{D}]$ or $\mathcal{D}^{\mathcal{C}}$. In particular $\mathbf{sSet} = \mathbf{Set}^{\Delta^{\mathrm{op}}}$ is a category whose morphisms are natural transformations of presheaves.

Definition 7.27 (Adjunction, hom-set form)**.** Functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ form an **adjoint pair** (F **left adjoint to** G, written $F \dashv G$) when there exists a family of bijections

$$\phi_{A,B} : \hom_{\mathcal{D}}(FA, B) \xrightarrow{\cong} \hom_{\mathcal{C}}(A, GB)$$

indexed by (A, B), **natural in** A **and** B: for every $f : A' \to A$ in $\mathcal{C}$, $g : B \to B'$ in $\mathcal{D}$, and $h : FA \to B$,

$$\phi_{A',B'}(g \circ h \circ Ff) = Gg \circ \phi_{A,B}(h) \circ f.$$

The next theorem gives the equivalent presentation that the Spivak adjunction (§7.5.4) uses directly.

Theorem 7.28 (Adjunction, unit-counit form)**.** $F \dashv G$ in the sense of Definition 7.27 if and only if there exist natural transformations

$$\eta : 1_{\mathcal{C}} \Rightarrow GF \quad (\textbf{unit}), \qquad \varepsilon : FG \Rightarrow 1_{\mathcal{D}} \quad (\textbf{counit}),$$

satisfying the **triangle identities**

$$\varepsilon_{FA} \circ F\eta_A = 1_{FA} \text{ in } \mathcal{D}, \qquad G\varepsilon_B \circ \eta_{GB} = 1_{GB} \text{ in } \mathcal{C},$$

for every $A \in \mathrm{Ob}(\mathcal{C})$, $B \in \mathrm{Ob}(\mathcal{D})$.

Proof of Theorem 7.28

We prove the two directions in turn.

(Hom-set form ⇒ unit-counit form). Given the natural bijections ϕ, define

$$\eta_A \ := \ \phi_{A,FA}(1_{FA}) \ \in \ \hom_{\mathcal{C}}(A, GFA), \qquad \varepsilon_B \ := \ \phi^{-1}_{GB,B}(1_{GB}) \ \in \ \hom_{\mathcal{D}}(FGB, B).$$

Naturality of η. For $f : A' \to A$, set $h = 1_{FA}$ in the naturality identity of Definition 7.27 with $g = 1_{FA}$ (and the identity $1_{FA} \circ 1_{FA} \circ Ff = Ff$):

$$\phi_{A',FA}(Ff) \ = \ G1_{FA} \circ \phi_{A,FA}(1_{FA}) \circ f \ = \ \eta_A \circ f.$$

Now apply the naturality identity with $A' = A'$, $B = FA'$, $B' = FA$, $g = Ff$, and $h = 1_{FA'}$:

$$\phi_{A',FA}(Ff \circ 1_{FA'}) \ = \ \phi_{A',FA}(Ff) \ = \ GFf \circ \phi_{A',FA'}(1_{FA'}) \circ 1_{A'} \ = \ GFf \circ \eta_{A'}.$$

Equating the two expressions for $\phi_{A',FA}(Ff)$:

$$\eta_A \circ f \ = \ GFf \circ \eta_{A'},$$

the naturality square for $\eta : 1_{\mathcal{C}} \Rightarrow GF$.

Naturality of ε. For $g : B \to B'$ in $\mathcal{D}$, we show $\varepsilon_{B'} \circ FGg = g \circ \varepsilon_B$. Apply $\phi_{GB,B'}$ to each side. Naturality in the second argument (with $g' = g$, $h = \varepsilon_B$) gives

$$\phi_{GB,B'}(g \circ \varepsilon_B) \;=\; Gg \circ \phi_{GB,B}(\varepsilon_B) \;=\; Gg \circ 1_{GB} \;=\; Gg,$$

using $\phi_{GB,B}(\varepsilon_B) = 1_{GB}$. Naturality in the first argument (with $f = Gg$, starting hom $\varepsilon_{B'} \in \hom(FGB', B')$) gives

$$\phi_{GB,B'}(\varepsilon_{B'} \circ FGg) \;=\; \phi_{GB',B'}(\varepsilon_{B'}) \circ Gg \;=\; 1_{GB'} \circ Gg \;=\; Gg.$$

Both expressions equal Gg; bijectivity of $\phi_{GB,B'}$ gives $\varepsilon_{B'} \circ FGg = g \circ \varepsilon_B$.

Triangle identity 2: $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$. By definition $\phi_{GB,B}(\varepsilon_B) = 1_{GB}$. Naturality in the second argument with $g = \varepsilon_B$, $h = 1_{FGB}$ gives

$$\phi_{GB,B}(\varepsilon_B \circ 1_{FGB}) \;=\; G\varepsilon_B \circ \phi_{GB,FGB}(1_{FGB}) \;=\; G\varepsilon_B \circ \eta_{GB},$$

using $\phi_{GB,FGB}(1_{FGB}) = \eta_{GB}$. The left side is $\phi_{GB,B}(\varepsilon_B) = 1_{GB}$, hence $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$.

Triangle identity 1: $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$. Apply $\phi_{A,FA}$ to both sides; bijectivity reduces the claim to $\phi_{A,FA}(\varepsilon_{FA} \circ F\eta_A) = \phi_{A,FA}(1_{FA}) = \eta_A$. Naturality in the second argument (with $g = \varepsilon_{FA}$, $h = F\eta_A$) gives

$$\phi_{A,FA}(\varepsilon_{FA} \circ F\eta_A) \;=\; G\varepsilon_{FA} \circ \phi_{A,FGFA}(F\eta_A).$$

Naturality in the first argument (with $f = \eta_A$, starting hom 1_{FGFA}) gives

$$\phi_{A,FGFA}(F\eta_A) \;=\; \phi_{A,FGFA}(1_{FGFA} \circ F\eta_A) \;=\; \phi_{GFA,FGFA}(1_{FGFA}) \circ \eta_A \;=\; \eta_{GFA} \circ \eta_A.$$

Substitute and apply Triangle identity 2 at $B = FA$:

$$\phi_{A,FA}(\varepsilon_{FA} \circ F\eta_A) \;=\; G\varepsilon_{FA} \circ \eta_{GFA} \circ \eta_A \;=\; 1_{GFA} \circ \eta_A \;=\; \eta_A.$$

(Unit-counit form $\Rightarrow$ hom-set form). Given (η, ε) satisfying the triangle identities, define

$$\phi_{A,B} : \hom_{\mathcal{D}}(FA, B) \to \hom_{\mathcal{C}}(A, GB), \qquad h \mapsto Gh \circ \eta_A,$$

$$\psi_{A,B} : \hom_{\mathcal{C}}(A, GB) \to \hom_{\mathcal{D}}(FA, B), \qquad k \mapsto \varepsilon_B \circ Fk.$$

ψ **inverts** ϕ**.** For $h : FA \to B$,

$$\begin{aligned}\psi(\phi(h)) &= \varepsilon_B \circ F(Gh \circ \eta_A) = \varepsilon_B \circ FGh \circ F\eta_A \\ &\overset{\text{(nat. }\varepsilon)}{=} h \circ \varepsilon_{FA} \circ F\eta_A \overset{\text{(triangle 1)}}{=} h \circ 1_{FA} = h.\end{aligned}$$

For $k : A \to GB$,

$$\begin{aligned}\phi(\psi(k)) &= G(\varepsilon_B \circ Fk) \circ \eta_A = G\varepsilon_B \circ GFk \circ \eta_A \\ &\overset{\text{(nat. }\eta)}{=} G\varepsilon_B \circ \eta_{GB} \circ k \overset{\text{(triangle 2)}}{=} 1_{GB} \circ k = k.\end{aligned}$$

ϕ and ψ are mutually inverse.

Naturality of ϕ**.** For $f : A' \to A$, $g : B \to B'$, $h : FA \to B$,

$$\begin{aligned}\phi_{A',B'}(g \circ h \circ Ff) &= G(g \circ h \circ Ff) \circ \eta_{A'} \\ &= Gg \circ Gh \circ GFf \circ \eta_{A'} \\ &\overset{\text{(nat. }\eta)}{=} Gg \circ Gh \circ \eta_A \circ f \\ &= Gg \circ \phi_{A,B}(h) \circ f. \qquad \blacksquare\end{aligned}$$

Example 7.4 (Free-forgetful adjunction)**.** For $F : \mathbf{Set} \to \mathbf{Vec}_{\mathbb{R}}$ and $U : \mathbf{Vec}_{\mathbb{R}} \to \mathbf{Set}$ as in Example 7.3(ii)–(iii), $F \dashv U$. The hom-set bijection

$$\phi_{A,V} : \hom_{\mathbf{Vec}}(\mathbb{R}^A, V) \xrightarrow{\cong} \hom_{\mathbf{Set}}(A, UV)$$

sends a linear map T to its restriction $a \mapsto T(e_a)$ on the basis $\{e_a\}$, with inverse the linear extension. The unit $\eta_A : A \to UFA = U(\mathbb{R}^A)$ is $a \mapsto e_a$; the counit $\varepsilon_V : FUV \to V$ is the linear extension of id_V, summing formal $\mathbb{R}$-combinations.

Adjoints determine each other

A functor admits at most one right adjoint up to natural isomorphism (and likewise for left). The Spivak adjunction (§7.5.4) is therefore canonical: given the geometric realization $|\cdot| : \mathbf{sFuzz} \to \mathbf{FinEPMet}$, its right adjoint Sing is determined up to natural isomorphism. UMAP exploits this canonicity to declare its optimization target unambiguously.

1.8 Fuzzy sets and fuzzy simplicial sets

UMAP replaces the binary "edge present / absent" of a graph with a strength of membership in $(0,1]$. The categorical machinery of §1.7 extends to this setting: replace **Set** with the category of fuzzy sets, then take presheaves on Δ valued in fuzzy sets.

Definition 7.29 (Fuzzy set)**.** A **fuzzy set** on a (non-fuzzy) ground set A is a function $\mu : A \to [0,1]$ assigning to each $a \in A$ a **strength of membership** $\mu(a)$. The **support** is $\operatorname{supp}(\mu) := \{a : \mu(a) > 0\}$. A fuzzy set is **strict** when $\mu : A \to (0,1]$ (every element has positive strength).

Definition 7.30 (Category **Fuzz**)**. Fuzz** has:

- objects: pairs (A, μ) with A a set and $\mu : A \to [0,1]$;
- morphisms $(A,\mu) \to (B,\nu)$: functions $f : A \to B$ such that $\mu(a) \leqslant \nu(f(a))$ for every $a \in A$;
- composition: composition of underlying functions; identities: identity functions.

The condition $\mu(a) \leqslant \nu(f(a))$ records that membership is preserved (or strengthened) along f.

Associativity and unit laws follow from those of **Set**.

A second presentation reuses the unit interval as a poset.

Definition 7.31 (Unit interval as a category)**.** $\mathbb{I} := (0, 1]$ is a poset under $\leqslant$. Treated as a category $\mathcal{I}$ via Example 7.2(iv), $\mathcal{I}$ has objects $a \in (0, 1]$ and a unique morphism $a \to b$ when $a \leqslant b$. The opposite category $\mathcal{I}^{\mathrm{op}}$ has the unique morphism $b \to a$ when $a \leqslant b$.

Lemma 7.32 (Fuzzy sets as $\mathcal{I}^{\mathrm{op}}$-presheaves; finite ground sets)**.** The category $\mathbf{Fuzz}^{\mathrm{fin}}$ of strict fuzzy sets on finite ground sets is equivalent to the full subcategory of $\mathbf{Set}^{\mathcal{I}^{\mathrm{op}}}$ whose objects are presheaves $X : \mathcal{I}^{\mathrm{op}} \to \mathbf{Set}$ taking values in finite sets, with every restriction map $X(b) \to X(a)$ (for $a \leqslant b$) an injection.

Proof. Object correspondence. Given a strict fuzzy set $\mu : A \to (0, 1]$ on finite A, define $X_\mu(a) := \{x \in A : \mu(x) \geqslant a\}$ for $a \in (0, 1]$. Each $X_\mu(a)$ is a finite subset of A, and for $a \leqslant b$ the inclusion $X_\mu(b) \hookrightarrow X_\mu(a)$ is the (manifestly injective) restriction map of the resulting presheaf. Conversely, given an injective presheaf X valued in finite sets, take A to be the eventual stable value of $X(a)$ as $a \to 0^+$; this set is finite by the assumption that each $X(a)$ is finite, and the chain has finitely many distinct values since $|X(a)|$ is monotone non-increasing in a and integer-valued. For $x \in A$, the set $\{a \in (0,1] : x \in X(a)\}$ is downward-closed, takes finitely many values, and admits a maximum (the chain stabilizes at finite steps), so $\mu_X(x) := \max\{a : x \in X(a)\}$ is attained. Then $X(a) = \{x : \mu_X(x) \geqslant a\}$ holds exactly for every $a \in (0,1]$.

Morphism correspondence. A morphism of fuzzy sets $f : (A, \mu) \to (B, \nu)$ satisfies $\mu(a) \leqslant \nu(f(a))$ iff for every threshold α, $X_\mu(\alpha) \subseteq f^{-1}(X_\nu(\alpha))$, equivalently f restricts to a function $X_\mu(\alpha) \to X_\nu(\alpha)$ commuting with the inclusion structure as α varies. This is exactly a natural transformation between the corresponding presheaves.

The two assignments are mutually inverse on objects and on morphisms. □

The lemma lets the categorical formalism handle fuzzy sets uniformly with simplicial sets. The next definition combines the two.

Definition 7.33 (Fuzzy simplicial set)**.** A **fuzzy simplicial set** is a functor

$$X : \Delta^{\mathrm{op}} \times \mathcal{I}^{\mathrm{op}} \to \mathbf{Set}$$

such that for each $[n] \in \Delta$, the restriction $X([n], -) : \mathcal{I}^{\mathrm{op}} \to \mathbf{Set}$ is an injective presheaf in the sense of Lemma 7.32. Equivalently, X is a functor $\Delta^{\mathrm{op}} \to \mathbf{Fuzz}$. The category of fuzzy simplicial sets and natural transformations is denoted **sFuzz**.
An element of $X([n], a)$ is an **n-simplex of strength** a; injectivity in $\mathcal{I}^{\mathrm{op}}$ means that an n-simplex of strength b has membership $\geqslant a$ for every $a \leqslant b$ via the restriction map.

A fuzzy simplicial set is therefore a system of simplicial sets $X^a := X(-, a) \in \mathbf{sSet}$ for $a \in (0, 1]$, with inclusions $X^b \hookrightarrow X^a$ whenever $a \leqslant b$ (stronger simplices are weaker simplices too).

The objects on the metric side come next.

Definition 7.34 (Extended-pseudo-metric space)**.** An **extended-pseudo-metric space** is a pair (X, d) with X a set and $d : X \times X \to [0, \infty]$ satisfying, for every $x, y, z \in X$:

(i) $d(x, x) = 0$ (reflexivity);

(ii) $d(x, y) = d(y, x)$ (symmetry);

(iii) $d(x, z) \leqslant d(x, y) + d(y, z)$ (triangle inequality, with $\infty + a := \infty$).

The ”pseudo” qualifier drops the implication $d(x, y) = 0 \Rightarrow x = y$; the ”extended” qualifier admits

$d(x, y) = \infty$.

Definition 7.35 (Category **FinEPMet**). **FinEPMet** has:

- objects: extended-pseudo-metric spaces (X, d) with X a finite set;
- morphisms $(X, d_X) \to (Y, d_Y)$: short maps, $d_Y(f(x), f(x')) \leqslant d_X(x, x')$ for every x, x';
- composition: composition of functions; identities: identity functions.

The two categories **sFuzz** and **FinEPMet** are linked by an adjunction (Spivak, 2009), used by UMAP. The full statement and proof appear in §7.5.4 once the manifold-modeling motivation is in place.

2 Principal Component Analysis

2.1 The two PCA problems

Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with rows $x_1^\top, \ldots, x_n^\top$, centered so that $\bar{x} = \mathbf{0}$ (achieved by left-multiplying by the centering matrix $\mathbf{M}_0$ of §1.4). The sample covariance is $S = \frac{1}{n-1}X^\top X$.

Definition 7.36 (Variance-maximization formulation)**.** The ***k*-th principal direction** is the unit vector $u_k \in \mathbb{R}^p$ solving

$$u_k = \arg\max_{\substack{u\in\mathbb{R}^p,\ \|u\|=1 \\ u\perp u_1,\dots,u_{k-1}}} u^\top S u, \qquad k = 1, 2, \dots, p.$$

The ***k*-th principal score** of observation i is $z_{ik} = u_k^\top x_i$. The directions u_k are the **loadings**; the scores are the coordinates of the data in the principal frame.

The objective $u^\top S u$ equals the sample variance of the projections $\{u^\top x_i\}_{i=1}^n$ when X is centered:

$$\frac{1}{n-1}\sum_{i=1}^n (u^\top x_i)^2 = u^\top\Big(\frac{1}{n-1}X^\top X\Big)u = u^\top S u.$$

Definition 7.37 (Reconstruction-error formulation)**.** For $k \leqslant p$, let $V \in \mathbb{R}^{p\times k}$ have orthonormal columns spanning a subspace $\mathcal{V} \subseteq \mathbb{R}^p$. The orthogonal projection of x_i onto $\mathcal{V}$ is $VV^\top x_i$, with reconstruction error $\|x_i - VV^\top x_i\|^2$. The **rank-k PCA subspace** is the $\mathcal{V}$ minimizing

$$R_k(\mathcal{V}) \;=\; \sum_{i=1}^{n} \|x_i - VV^\top x_i\|^2 \;=\; \|X - XVV^\top\|_F^2.$$

2.2 Variance-maximization solution via Rayleigh

Theorem 7.38 (Principal directions are eigenvectors of S)**.** Let $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$ be the eigenvalues of S with orthonormal eigenvectors $q_1, \ldots, q_p$ (Theorem 7.2). The choice $u_k := q_k$ satisfies Definition 7.36, with maximized objective

$$u_k^\top S u_k \;=\; \lambda_k.$$

Proof. $S = \frac{1}{n-1} X^\top X$ is symmetric, and $v^\top S v = \frac{1}{n-1}\|Xv\|^2 \geqslant 0$, so PSD; the eigenvalues are real and nonnegative.

Show $u_k = q_k$ satisfies Definition 7.36 by induction on k. Unit 2 Theorem 2.7 with $A = S$ states

$$\lambda_k \;=\; \max\{u^\top S u : \|u\| = 1,\; u \perp q_1, \ldots, q_{k-1}\},$$

attained at q_k. Definition 7.36 differs only by replacing $q_1, \ldots, q_{k-1}$ in the orthogonality constraint with the inductively-defined $u_1, \ldots, u_{k-1}$. The base case $k = 1$ has empty constraint in both, so $u_1 = q_1$ with maximum λ_1. The inductive step substitutes $u_j = q_j$ for $j < k$ (hypothesis), making the two constraint sets identical; $u_k = q_k$ remains a valid choice with $u_k^\top S u_k = q_k^\top S q_k = \lambda_k$. □

Lemma 7.39 (Variance decomposition)**.** Total sample variance $\operatorname{tr}(S)$ decomposes as

$$\operatorname{tr}(S) = \sum_{k=1}^{p} \lambda_k = \sum_{k=1}^{p} u_k^\top S u_k.$$

The first k principal directions capture $\sum_{j=1}^{k} \lambda_j$ of the total.

Proof. $\operatorname{tr}(S) = \operatorname{tr}(Q\Lambda Q^\top) = \operatorname{tr}(\Lambda Q^\top Q) = \operatorname{tr}(\Lambda) = \sum_k \lambda_k$ by cyclicity of the trace and $Q^\top Q = I_p$. The second equality is Theorem 7.38. □

2.3 Reconstruction-error formulation and equivalence

Theorem 7.40 (Equivalence of variance-max and reconstruction-min)**.** With S, λ_j, q_j as in Theorem 7.38,

$$\mathcal{V}^\star \;=\; \operatorname{span}(q_1, \dots, q_k)$$

minimizes the rank-k reconstruction error of Definition 7.37, with minimum value

$$R_k(\mathcal{V}^\star) \;=\; (n-1)\sum_{j>k} \lambda_j.$$

Proof. The singular values of X satisfy $\sigma_j^2 = \lambda_j(X^\top X) = (n-1)\lambda_j$ (Theorem 7.5 and $X^\top X = (n-1)S$). Set $Q_k = [q_1 \;\cdots\; q_k] \in \mathbb{R}^{p\times k}$; $Q_kQ_k^\top$ is the orthogonal projection onto $\mathcal{V}^\star$.
Step 1: Reconstruction error at $\mathcal{V}^\star$. With $\|M\|_F^2 = \operatorname{tr}(M^\top M)$ and $Q_kQ_k^\top$ symmetric idempotent,

$$\|X - XQ_kQ_k^\top\|_F^2 \;=\; \operatorname{tr}\left((I - Q_kQ_k^\top)X^\top X(I - Q_kQ_k^\top)\right) \;=\; \operatorname{tr}\left(X^\top X(I - Q_kQ_k^\top)\right),$$

the second equality by trace cyclicity and $(I - Q_kQ_k^\top)^2 = I - Q_kQ_k^\top$. Using the outer-product form $X^\top X = (n-1)\sum_j \lambda_j q_j q_j^\top$ and $\operatorname{tr}(uv^\top) = v^\top u$,

$$\operatorname{tr}\left(X^\top X(I - Q_k Q_k^\top)\right) \;=\; (n-1)\sum_{j=1}^{p} \lambda_j\, q_j^\top (I - Q_k Q_k^\top) q_j.$$

The vector $Q_k^\top q_j$ has i-th entry $q_i^\top q_j = \delta_{ij}$ for $i \leqslant k$, so $q_j^\top Q_k Q_k^\top q_j = \|Q_k^\top q_j\|^2$ equals 1 if $j \leqslant k$ and 0 if $j > k$. Hence $q_j^\top (I - Q_k Q_k^\top) q_j = \mathbb{1}[j > k]$, and

$$\|X - XQ_kQ_k^\top\|_F^2 \;=\; (n-1)\sum_{j>k} \lambda_j.$$

Step 2: Eckart–Young lower bound. For any orthonormal $V \in \mathbb{R}^{p\times k}$ spanning a subspace $\mathcal{V}$, the matrix $XVV^\top$ has rank at most k. By Theorem 7.8,

$$R_k(\mathcal{V}) \;=\; \|X - XVV^\top\|_F^2 \;\geqslant\; \min_{\operatorname{rank} B \leqslant k} \|X - B\|_F^2 \;=\; \sum_{j>k} \sigma_j^2 \;=\; (n-1)\sum_{j>k} \lambda_j.$$

Step 3: $\mathcal{V}^\star$ achieves the bound. Steps 1 and 2 give $R_k(\mathcal{V}^\star) = (n-1)\sum_{j>k} \lambda_j$, matching the lower bound; hence $\mathcal{V}^\star$ minimizes R_k. $\square$

2.4 The SVD perspective

PCA on centered X can be computed without forming S: the singular value decomposition of X already exposes the principal directions as the right singular vectors and the principal scores as XV.

Theorem 7.41 (PCA via SVD). Let $X \in \mathbb{R}^{n\times p}$ be centered with reduced SVD $X = U\Sigma V^\top$, where $U \in \mathbb{R}^{n\times r}$, $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $V \in \mathbb{R}^{p\times r}$, and $r = \operatorname{rank} X$. Then:

(i) The columns v_j of V are orthonormal eigenvectors of S at eigenvalues $\lambda_j = \sigma_j^2/(n-1)$.

(ii) The principal-score matrix $Z := XV \in \mathbb{R}^{n\times r}$ equals $U\Sigma$, with entries $z_{ij} = \sigma_j U_{ij}$.

(iii) The rank-k truncated SVD satisfies $X_k = XV_kV_k^\top$; its column span $\operatorname{span}(v_1, \ldots, v_k)$ is a top-k eigenspace of S, hence minimizes the rank-k reconstruction error (Theorem 7.40).

Proof. **(i)** $X^\top X = V\Sigma^\top\Sigma V^\top = V \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2)V^\top$, so each v_j satisfies $X^\top X v_j = \sigma_j^2 v_j$, equivalently $Sv_j = (\sigma_j^2/(n-1))v_j$. Orthonormality of $\{v_j\}$ is part of the SVD definition.
(ii) $XV = U\Sigma V^\top V = U\Sigma$, using $V^\top V = I_r$. Entry-by-entry, $z_{ij} = (U\Sigma)_{ij} = \sigma_j U_{ij}$.
(iii) $XV_kV_k^\top = U\Sigma V^\top V_kV_k^\top$. Since V_k takes the first k orthonormal columns of V, $V^\top V_k =$

$\left(\begin{smallmatrix} I_k \\ \mathbf{0}_{(r-k)\times k} \end{smallmatrix}\right)$, so $\Sigma V^\top V_k = \left(\begin{smallmatrix} \Sigma_k \\ \mathbf{0} \end{smallmatrix}\right)$, and

$$XV_kV_k^\top = U\begin{pmatrix} \Sigma_k \\ \mathbf{0} \end{pmatrix} V_k^\top = U_k\Sigma_kV_k^\top = X_k.$$

By (i), $\{v_1, \ldots, v_k\}$ are orthonormal eigenvectors of S at the top-k eigenvalues $\lambda_1, \ldots, \lambda_k$, so $\operatorname{span}(v_1, \ldots, v_k)$ is a top-k eigenspace of S. □

Numerical practice

For $p \gg n$ (gene-expression-style data) or moderately conditioned X, SVD on X avoids forming the $p \times p$ matrix $S = \frac{1}{n-1}X^\top X$. Squaring X amplifies any conditioning issue, since $\kappa(X^\top X) = \kappa(X)^2$, so eigendecomposing S loses precision faster than SVD on X. The command `prcomp(X, center = TRUE)` computes PCA via SVD internally; `princomp(X)` eigendecomposes S and is now legacy.

2.5 Loadings, scores, biplot

PCA produces two output matrices: the loadings $V = [u_1 \ \cdots \ u_p]$ (principal directions in feature space) and the scores $Z = XV = U\Sigma$ (coordinates in the principal frame, Theorem 7.41).

Loadings. Column j of V is a vector in feature space; its entries $v_{m,j}$ are the weights of the original features in the j-th principal score, $z_{ij} = \sum_m v_{m,j} x_{i,m}$. A large $|v_{m,j}|$ identifies feature m as a major contributor to component j. The sign of each v_j is arbitrary: replacing v_j by $-v_j$ flips all z_{ij} to $-z_{ij}$ without affecting variance or reconstruction.

Scores. Row i of Z is a vector in $\mathbb{R}^r$; its first k entries are the coordinates of observation i in the rank-k PCA subspace $\mathcal{V}^\star$. The reconstruction $\hat{x}_i^{(k)} := \sum_{j=1}^k z_{ij} v_j$ is the orthogonal projection of x_i onto $\mathcal{V}^\star$ (Theorem 7.40); equivalently, the rows of $X_k = XV_kV_k^\top$ are the $\hat{x}_i^{(k)}$.

Biplot. A two-dimensional **biplot** (Gabriel, 1971) superimposes two scatter plots on the same axes: the scores $\{(z_{i,1}, z_{i,2}) : i = 1, \ldots, n\}$ as points, and the loadings $\{(v_{m,1}, v_{m,2}) : m = 1, \ldots, p\}$ as arrows from the origin (often rescaled for visibility). Nearby points are observations with similar component-1 and component-2 coordinates; arrows pointing in similar directions are features that load similarly on those two components.

2.6 Variance explained, scree plot, choice of k

Definition 7.42 (Fraction of variance explained)**.** For PCA with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$ of S, the **fraction of variance explained** by the first k principal directions is

$$\mathrm{FVE}_k \;:=\; \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{p} \lambda_j} \;=\; \frac{\sum_{j=1}^{k} \lambda_j}{\operatorname{tr}(S)} \;\in\; [0,1].$$

The **cumulative variance curve** plots FVE_k against k; the **scree plot** plots λ_j against j.

By Lemma 7.39 and Theorem 7.40,

$$1 - \mathrm{FVE}_k \;=\; \frac{\sum_{j>k} \lambda_j}{\operatorname{tr}(S)} \;=\; \frac{R_k(\mathcal{V}^\star)}{(n-1)\operatorname{tr}(S)}.$$

FVE_k has two readings: the variance fraction captured (left) and the complement of the rank-k reconstruction error normalized by total variance (right).

Choosing k.

(1) **Cumulative variance threshold.** Pick the smallest k with $\mathrm{FVE}_k \geqslant 0.9$ (or 0.8, 0.95,

problem-dependent).

(2) **Scree elbow.** Plot λ_j vs. j and pick k at the visual "elbow" where the curve flattens. The elbow exists because λ_j decreases rapidly for small j and stabilizes.

(3) **Kaiser rule (correlation PCA).** Keep components with $\lambda_j > 1$. The threshold is the mean eigenvalue of R, which equals 1 since $\mathrm{tr}(R) = p$; components above the threshold capture more variance than a single original (standardized) feature.

The three rules can disagree by several components. None has sharp theoretical justification; the Kaiser rule in particular has fallen out of favor (Jolliffe, 2002, §6.1.2) for over-retaining components on weakly-correlated data. Methods with statistical backing (parallel analysis, profile likelihood) replace the visual elbow with a calibrated comparison against random-data baselines.

2.7 Standardization: PCA on covariance vs correlation

The PCA problems of §2.1 were stated for centered data X. A second preprocessing choice (whether to scale each feature to unit sample variance) produces two different PCA outputs.

Definition 7.43 (Standardized data, correlation matrix)**.** Let X be centered, with sample covariance S and per-feature sample standard deviations $s_j = \sqrt{S_{jj}}$. The **standardized data matrix** is $\widetilde{X} := XD^{-1}$ where $D = \operatorname{diag}(s_1, \dots, s_p)$. The **sample correlation matrix** is

$$R := \frac{1}{n-1}\widetilde{X}^\top \widetilde{X} = D^{-1}SD^{-1},$$

with entries $R_{jk} = S_{jk}/(s_j s_k) \in [-1, 1]$ and $R_{jj} = 1$.

PCA on X uses eigenvectors of S; PCA on $\widetilde{X}$ uses eigenvectors of R. Unless $D = cI_p$ (all features have equal sample variance), the two eigenvector sets differ and are not related by a closed-form transformation: $R = D^{-1}SD^{-1}$ is a similarity of S only when D is scalar, and a non-scalar D^{-1} rotates the basis non-orthogonally.

When each choice is appropriate.

(1) **Covariance PCA** preserves the original units. Appropriate when all features are on a common scale and variance differences between features are scientifically meaningful.

(2) **Correlation PCA** eliminates scale differences. Appropriate when features are in different units

(height in cm, weight in kg) or when sample variances are artifacts of the recording scale.

3 K-means and Hierarchical Clustering

3.1 K-means objective and Lloyd's algorithm

K-means partitions n observations into K clusters by minimizing the within-cluster sum of squared distances to cluster centers.

Definition 7.44 (K-means objective)**.** For $X \in \mathbb{R}^{n\times p}$ with rows $x_1, \ldots, x_n$ and a fixed integer $K \in \{1, \ldots, n\}$, the **K-means objective** jointly minimizes over a partition $\mathcal{C} = (C_1, \ldots, C_K)$ of $\{1, \ldots, n\}$ into K non-empty parts and cluster centers $\mu_1, \ldots, \mu_K \in \mathbb{R}^p$:

$$J(\mathcal{C}, \boldsymbol{\mu}) = \sum_{k=1}^{K} \sum_{i\in C_k} \|x_i - \mu_k\|^2.$$

For fixed partition $\mathcal{C}$, the inner minimization over μ_k is unconstrained least-squares with closed-form solution $\mu_k = \bar{x}^{(k)} := \frac{1}{|C_k|}\sum_{i\in C_k} x_i$, the cluster centroid. The minimum over centers, given $\mathcal{C}$, is

$$J^\star(\mathcal{C}) = \sum_{k=1}^{K}\sum_{i\in C_k}\|x_i - \bar{x}^{(k)}\|^2,$$

the **within-cluster sum of squares** (WCSS). The K-means problem is therefore the discrete optimization $\min_{\mathcal{C}} J^\star(\mathcal{C})$ over all partitions of $\{1,\dots,n\}$ into K non-empty parts.

Lemma 7.45 (Total sum-of-squares decomposition)**.** With $\bar{x} = \frac{1}{n}\sum_i x_i$ the global mean and $n_k = |C_k|$,

$$\underbrace{\sum_{i=1}^{n}\|x_i - \bar{x}\|^2}_{\text{TSS}} = \underbrace{\sum_{k=1}^{K}\sum_{i\in C_k}\|x_i - \bar{x}^{(k)}\|^2}_{J^\star(\mathcal{C}) = \text{WCSS}} + \underbrace{\sum_{k=1}^{K} n_k\|\bar{x}^{(k)} - \bar{x}\|^2}_{\text{BCSS}}.$$

Proof. For each C_k, expand $\|x_i - \bar{x}\|^2 = \|x_i - \bar{x}^{(k)}\|^2 + 2(x_i - \bar{x}^{(k)})^\top(\bar{x}^{(k)} - \bar{x}) + \|\bar{x}^{(k)} - \bar{x}\|^2$. Sum over $i \in C_k$; the cross term vanishes by the centroid identity $\sum_{i\in C_k}(x_i - \bar{x}^{(k)}) = \mathbf{0}$. Summing the resulting per-cluster identity over k gives the claim. □

TSS is fixed by the data, so minimizing WCSS is equivalent to maximizing BCSS, the spread of cluster centroids around the global mean.

The discrete optimization $\min_{\mathcal{C}} J^\star(\mathcal{C})$ is NP-hard in general. Lloyd's algorithm (Lloyd, 1982), described in earlier unpublished form and often jointly attributed to Forgy (1965), gives an iterative heuristic that converges to a local minimum.

Definition 7.46 (Lloyd's algorithm)**.** Given initial centers $\mu_1^{(0)}, \ldots, \mu_K^{(0)} \in \mathbb{R}^p$, iterate for $t = 0, 1, 2, \ldots$:

- **Assignment step.** Each observation joins the cluster of its nearest center:

$$C_k^{(t)} = \{i : k = \arg\min_j \|x_i - \mu_j^{(t)}\|^2\}, \qquad k = 1, \ldots, K$$

(ties broken by lowest index).

- **Update step.** Each center moves to the centroid of its cluster:

$$\mu_k^{(t+1)} = \frac{1}{|C_k^{(t)}|} \sum_{i \in C_k^{(t)}} x_i, \qquad k = 1, \dots, K.$$

Stop when $\mathcal{C}^{(t+1)} = \mathcal{C}^{(t)}$.

Theorem 7.47 (Monotone descent and finite-time convergence)**.** Lloyd's algorithm produces a sequence $J(\mathcal{C}^{(t)}, \boldsymbol{\mu}^{(t)})$ that is monotone non-increasing in t, and the algorithm halts after finitely many iterations at a partition $\mathcal{C}^\star$ for which no single observation can be reassigned to a different cluster without increasing the objective.

Proof. **Monotonicity.** Write $J_t = J(\mathcal{C}^{(t)}, \boldsymbol{\mu}^{(t)})$. The update step replaces $\boldsymbol{\mu}^{(t)}$ by the centroids $\boldsymbol{\mu}^{(t+1)}$ of $\mathcal{C}^{(t)}$. With $\mathcal{C}^{(t)}$ fixed, each μ_k minimizes $\sum_{i \in C_k^{(t)}} \|x_i - \mu_k\|^2$ at the centroid (least-squares, unique minimizer), so $J(\mathcal{C}^{(t)}, \boldsymbol{\mu}^{(t+1)}) \leqslant J_t$. The assignment step at iteration $t+1$ produces $\mathcal{C}^{(t+1)}$ in which each i is assigned to its nearest center in $\boldsymbol{\mu}^{(t+1)}$. With $\boldsymbol{\mu}^{(t+1)}$ fixed, each observation contributes $\|x_i - \mu_{c(i)}\|^2$ where $c(i)$ is its cluster index, and the assignment rule chooses $c(i)$ to minimize this contribution per observation. Hence $J(\mathcal{C}^{(t+1)}, \boldsymbol{\mu}^{(t+1)}) \leqslant J(\mathcal{C}^{(t)}, \boldsymbol{\mu}^{(t+1)}) \leqslant J_t$, giving $J_{t+1} \leqslant J_t$.

Finite-time convergence. The number of K-part partitions of $\{1, \dots, n\}$ is at most $K^n < \infty$. We claim $J_{t+1} < J_t$ strictly whenever $\mathcal{C}^{(t+1)} \neq \mathcal{C}^{(t)}$: the assignment step assigns each observation to its argmin-distance cluster with ties broken uniquely, so a change in $\mathcal{C}$ implies some observation has moved to a strictly closer center under $\boldsymbol{\mu}^{(t+1)}$, lowering its squared-distance contribution. Since $\mathcal{C}^{(t)}$ takes values in a finite set and the objective strictly decreases at every change, the sequence stabilizes at some $t^\star$ with $\mathcal{C}^{(t^\star+1)} = \mathcal{C}^{(t^\star)}$ and the algorithm halts.

Local optimality. At halting, every i is at its nearest centroid. Reassigning a single i from $C_{c(i)}^{(t^\star)}$ to $C_j^{(t^\star)}$ replaces $\|x_i - \bar{x}^{(c(i))}\|^2$ by $\|x_i - \bar{x}^{(j)}\|^2 \geqslant \|x_i - \bar{x}^{(c(i))}\|^2$, so J does not decrease. $\square$

Local optima are not global

Different initializations $\boldsymbol{\mu}^{(0)}$ produce different local optima with sometimes substantially different objective values. The standard remedy in R is `kmeans(x, centers, nstart = 25)`, which runs Lloyd's algorithm 25 times from random initializations and returns the best.

3.2 K-means++ initialization

Random uniform initialization (each $\mu_k^{(0)}$ a random row of X) often produces poor configurations: two centers in the same true cluster, leaving another true cluster without a center. K-means++ (Arthur and Vassilvitskii, 2007) replaces uniform initialization with a D^2-weighted scheme that systematically spreads initial centers across the data.

Definition 7.48 (K-means++ initialization)**.** For data X and K centers, sample:

(1) $\mu_1^{(0)} := x_i$ for i chosen uniformly from $\{1, \ldots, n\}$;

(2) for $j = 2, \ldots, K$, sample $\mu_j^{(0)} := x_i$ from

$$\Pr(i \mid \mu_1^{(0)}, \ldots, \mu_{j-1}^{(0)}) \propto D(x_i)^2, \qquad D(x_i) := \min_{l<j} \|x_i - \mu_l^{(0)}\|.$$

After all K centers are chosen, run Lloyd's algorithm from this initialization.

The squared-distance weighting $D(x_i)^2$ makes points far from existing centers proportionally more likely to be selected next.

Theorem 7.49 (K-means++ approximation guarantee)**.** Let $J^\star_{\text{opt}}$ be the global minimum of the K-means objective. The expected K-means++ objective after Lloyd's algorithm satisfies

$$\mathbb{E}\big[J^\star(\mathcal{C}_{\text{K-means++}})\big] \;\leqslant\; 8(\ln K + 2)\, J^\star_{\text{opt}},$$

where the expectation is over the randomness in the K-means++ initialization (Arthur and Vassilvitskii, 2007, Theorem 1.1).

K-means++ is therefore $O(\log K)$-competitive with the global optimum in expectation. In R, K-means++ initialization is provided by `ClusterR::KMeans_rcpp(initializer = "kmeans++")`; `stats::kmeans` uses uniform random initialization, with `nstart` > 1 partially compensating.

3.3 EM viewpoint via spherical Gaussian mixtures

K-means coincides with the limiting form of the Expectation-Maximization (EM) algorithm applied to a Gaussian mixture model with isotropic covariances $\sigma^2 I_p$ as $\sigma^2 \to 0$.

Definition 7.50 (Spherical-equal-variance Gaussian mixture)**.** A **spherical-equal-variance Gaussian mixture** on $\mathbb{R}^p$ has parameters $\boldsymbol{\mu} = (\mu_1, \dots, \mu_K)$ and $\sigma^2 > 0$. Observations are independent with

$$x_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \sigma^2 I_p), \qquad Z_i \sim \mathrm{Uniform}\{1, \dots, K\}.$$

Theorem 7.51 (K-means as $\sigma^2 \to 0$ EM limit)**.** The EM algorithm for Definition 7.50, in the $\sigma^2 \to 0$ limit, reduces to Lloyd's algorithm on the same data.

Proof. The E-step computes the posterior responsibilities

$$\gamma_{ik} = \Pr(Z_i = k \mid x_i; \boldsymbol{\mu}, \sigma^2) = \frac{\exp(-\|x_i - \mu_k\|^2/(2\sigma^2))}{\sum_{l=1}^{K} \exp(-\|x_i - \mu_l\|^2/(2\sigma^2))}.$$

Let $k^\star(i) = \arg\min_l \|x_i - \mu_l\|^2$ (assume unique). As $\sigma^2 \to 0$, the term in the denominator with the smallest $\|x_i - \mu_l\|^2$ (namely $l = k^\star(i)$, whose exponent is the largest) dominates, and

$$\gamma_{ik} \to \mathbb{1}\{k = k^\star(i)\} \qquad (\sigma^2 \to 0).$$

The M-step maximizes the expected complete-data log-likelihood over $\boldsymbol{\mu}$, with closed-form solution

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \gamma_{ik}^{(t)} x_i}{\sum_{i=1}^n \gamma_{ik}^{(t)}}.$$

In the $\sigma^2 \to 0$ limit, $\gamma_{ik}^{(t)} = \mathbb{1}\{k = k^\star(i)\}$ collapses the M-step update to $\mu_k^{(t+1)} = (1/|C_k^{(t)}|) \sum_{i \in C_k^{(t)}} x_i$, the centroid of $C_k^{(t)} = \{i : k^\star(i) = k\}$. The combined E–M iteration is exactly Lloyd's algorithm (Definition 7.46). $\square$

The Gaussian-mixture viewpoint clarifies the assumptions K-means implicitly makes:

Modeling assumption	**Failure mode**
Spherical covariance $\sigma^2 I_p$	Elongated or correlated clusters split or merge
Equal σ^2 across clusters	Wide and tight clusters compete unfairly for boundary points
Isotropic Euclidean distance	Different feature scales bias the partition

The first two are addressed by full Gaussian mixture models (without the $\sigma^2 \to 0$ limit), the third by standardization (§2.7).

Spectral clustering for non-spherical clusters

Spectral clustering (Ng et al., 2002) sidesteps the spherical-cluster assumption by first embedding the data in $\mathbb{R}^K$ via the bottom-K eigenvectors of a graph Laplacian, then K-means clustering in that embedding. Two-moon and concentric-ring data, where K-means fails badly, are partitioned correctly. R: `kernlab::specc`.

3.4 Agglomerative hierarchical clustering

Hierarchical clustering produces a nested sequence of partitions, from n singletons up to one all-encompassing cluster, by iteratively merging the two closest clusters under a linkage function.

Definition 7.52 (Agglomerative hierarchical clustering)**.** For data $x_1, \ldots, x_n \in \mathbb{R}^p$ with pairwise distances $d_{ij} = \|x_i - x_j\|$ and a linkage function $D : 2^{[n]} \times 2^{[n]} \to [0, \infty)$ assigning a distance to any pair of disjoint subsets, the **agglomerative algorithm** runs:

(1) Initialize $\mathcal{P}^{(0)} = \{\{1\}, \{2\}, \ldots, \{n\}\}$ (singletons), merge height $h^{(0)} = 0$.

(2) For $t = 0, 1, \ldots, n-2$: identify the disjoint pair $(C_a, C_b) \in \mathcal{P}^{(t)} \times \mathcal{P}^{(t)}$ minimizing $D(C_a, C_b)$. Merge: $\mathcal{P}^{(t+1)} := (\mathcal{P}^{(t)} \setminus \{C_a, C_b\}) \cup \{C_a \cup C_b\}$, with merge height $h^{(t+1)} = D(C_a, C_b)$.

(3) After $n-1$ merges, $\mathcal{P}^{(n-1)} = \{\{1, \ldots, n\}\}$.

The sequence $(\mathcal{P}^{(t)}, h^{(t)})_{t=0}^{n-1}$ is the **hierarchy**.

Definition 7.53 (Dendrogram)**.** The **dendrogram** of a hierarchy is a binary tree with n leaves (the singletons) and $n-1$ internal nodes (the merges). Each internal node is placed at vertical coordinate equal to its merge height $h^{(t)}$; the two children are the merged clusters.

A horizontal cut at height $h^\star$ yields a partition: the connected components of the dendrogram below height $h^\star$. The cut height parameterizes granularity, replacing K-means' K with a continuous threshold.

3.5 Linkage criteria

Definition 7.54 (Standard linkages)**.** For disjoint clusters $C_a, C_b \subseteq \{1, \ldots, n\}$ with sizes n_a, n_b and centroids $\bar{x}^{(a)}, \bar{x}^{(b)}$:

Linkage	**Definition** $D(C_a, C_b)$
Single	$\min_{i \in C_a, j \in C_b} d_{ij}$
Complete	$\max_{i \in C_a, j \in C_b} d_{ij}$
Average	$\dfrac{1}{n_a n_b} \sum_{i \in C_a, j \in C_b} d_{ij}$
Ward	$\dfrac{n_a n_b}{n_a + n_b} \lVert \bar{x}^{(a)} - \bar{x}^{(b)} \rVert^2$

Single linkage merges clusters with the closest pair of points; the result tends toward chaining, where elongated clusters form along density gradients.

Complete linkage merges clusters with the smallest maximum pair distance; the result is compact, ball-shaped clusters.

Average linkage averages all cross-cluster pair distances; behavior is intermediate between single and complete.

Ward's linkage (Ward, 1963) measures the within-cluster sum-of-squares cost of merging C_a, C_b into one cluster. The increase in WCSS (Lemma 7.45) when merging is exactly $\frac{n_a n_b}{n_a + n_b}\|\bar{x}^{(a)} - \bar{x}^{(b)}\|^2$ (Everitt et al., 2011, §4.4). Ward's linkage produces compact, balanced clusters and is the default for Euclidean data.

Centroid linkage and inversions

A fifth linkage, $D(C_a, C_b) = \|\bar{x}^{(a)} - \bar{x}^{(b)}\|$ (centroid linkage), can produce **inversions** where $h^{(t+1)} < h^{(t)}$, breaking the dendrogram interpretation. Single, complete, average, and Ward linkages are all monotone: $h^{(t)} \leqslant h^{(t+1)}$ throughout, ensuring a valid dendrogram. R's `hclust(method = "centroid")` can produce inverted dendrograms; the documentation flags this.

3.6 Lance–Williams recurrence

The four standard linkages all satisfy a single parametric recurrence (Lance and Williams, 1967), allowing the distance from a newly-formed cluster to all other clusters to be computed without rescanning the underlying data.

Theorem 7.55 (Lance–Williams recurrence)**.** Let C_a, C_b be merged into $C_{ab} := C_a \cup C_b$, and let C_c be any other cluster. For each linkage of Definition 7.54, the merged distance is

$$D(C_{ab}, C_c) \;=\; \alpha_a D(C_a, C_c) + \alpha_b D(C_b, C_c) + \beta D(C_a, C_b) + \gamma |D(C_a, C_c) - D(C_b, C_c)|,$$

with parameters

Linkage	α_a	α_b	β	γ
Single	$1/2$	$1/2$	0	$-1/2$
Complete	$1/2$	$1/2$	0	$+1/2$
Average	$\dfrac{n_a}{n_a+n_b}$	$\dfrac{n_b}{n_a+n_b}$	0	0
Ward	$\dfrac{n_a+n_c}{n_a+n_b+n_c}$	$\dfrac{n_b+n_c}{n_a+n_b+n_c}$	$-\dfrac{n_c}{n_a+n_b+n_c}$	0

Proof of Theorem 7.55, single and Ward cases

Single linkage. $D(C_{ab}, C_c) = \min_{i\in C_a\cup C_b,\, j\in C_c} d_{ij} = \min\{D(C_a, C_c), D(C_b, C_c)\}$. Using $\min(u, v) = \frac{1}{2}(u+v) - \frac{1}{2}|u-v|$, the recurrence holds with $\alpha_a = \alpha_b = 1/2$, $\beta = 0$, $\gamma = -1/2$. Complete linkage is symmetric: $\max(u, v) = \frac{1}{2}(u+v) + \frac{1}{2}|u-v|$ gives $\gamma = +1/2$.

Ward linkage. By Definition 7.54,

$$D(C_{ab}, C_c) = \frac{(n_a+n_b)n_c}{n_a+n_b+n_c}\left\|\bar{x}^{(ab)} - \bar{x}^{(c)}\right\|^2.$$

The merged centroid is $\bar{x}^{(ab)} = (n_a\bar{x}^{(a)} + n_b\bar{x}^{(b)})/(n_a + n_b)$, so

$$\bar{x}^{(ab)} - \bar{x}^{(c)} = \frac{n_a(\bar{x}^{(a)} - \bar{x}^{(c)}) + n_b(\bar{x}^{(b)} - \bar{x}^{(c)})}{n_a + n_b}.$$

Set $u := \bar{x}^{(a)} - \bar{x}^{(c)}$, $v := \bar{x}^{(b)} - \bar{x}^{(c)}$. Expanding $\|n_a u + n_b v\|^2$ and using $u^\top v = \frac{1}{2}(\|u\|^2 + \|v\|^2 - \|u - v\|^2)$:

$$\begin{aligned}(n_a + n_b)^2\|\bar{x}^{(ab)} - \bar{x}^{(c)}\|^2 &= n_a^2\|u\|^2 + n_b^2\|v\|^2 + n_a n_b\big(\|u\|^2 + \|v\|^2 - \|u - v\|^2\big) \\ &= n_a(n_a + n_b)\|u\|^2 + n_b(n_a + n_b)\|v\|^2 - n_a n_b\|u - v\|^2.\end{aligned}$$

Using $u - v = \bar{x}^{(a)} - \bar{x}^{(b)}$, multiply both sides by $n_c/[(n_a + n_b)(n_a + n_b + n_c)]$. The LHS becomes $D(C_{ab}, C_c)$. The RHS three terms become, using $\|\bar{x}^{(a)} - \bar{x}^{(c)}\|^2 = \frac{n_a+n_c}{n_a n_c}D(C_a, C_c)$ and the analogous identities for $D(C_b, C_c)$ and $D(C_a, C_b)$:

$$\frac{n_a + n_c}{n_a + n_b + n_c}D(C_a, C_c) + \frac{n_b + n_c}{n_a + n_b + n_c}D(C_b, C_c) - \frac{n_c}{n_a + n_b + n_c}D(C_a, C_b),$$

matching the tabulated parameters with $\gamma = 0$. Average linkage is verified analogously, using $D(C_{ab}, C_c) = (n_a + n_b)^{-1}(n_a D(C_a, C_c) + n_b D(C_b, C_c))$ directly. ■

Computational consequences

The Lance–Williams recurrence allows the agglomerative algorithm (Definition 7.52) to maintain a $k \times k$ distance matrix among current clusters and update it in $O(k)$ at each merge: the row and column for the merged cluster are computed from the rows of C_a, C_b via the recurrence, then the matrix shrinks by one row and column. Total time $O(n^3)$, total space $O(n^2)$. The data matrix is consulted only once at initialization to compute the $n \times n$ pairwise distance matrix.

HDBSCAN: density-based modern alternative

Hierarchical clustering with the standard linkages produces a hierarchy regardless of cluster shape or density. Density-based methods, in particular **HDBSCAN** (Campello et al., 2013), replace the linkage with a density-reachability criterion (mutual reachability distance) and extract a flat partition by selecting the most stable clusters across density thresholds. HDBSCAN handles clusters of varying density and identifies low-density points as noise; standard hierarchical clustering does neither. R: `dbscan::hdbscan`.

4 t-SNE: Stochastic Neighbor Embedding

4.1 The dimensionality-reduction problem beyond linearity

PCA produces a linear embedding: each $x_i \in \mathbb{R}^p$ is sent to $V_k^\top x_i \in \mathbb{R}^k$ where V_k is fixed. Linear embedding cannot recover low-dimensional structure that lives on a curved manifold inside $\mathbb{R}^p$.

Example 7.5 (Swiss roll)**.** The Swiss-roll dataset is generated by drawing (t, h) with $t \in [3\pi/2,\ 9\pi/2]$ and $h \in [0, 21]$ and embedding into $\mathbb{R}^3$ via

$$x = (t\cos t,\ h,\ t\sin t).$$

The intrinsic two-dimensional structure (t, h) is recoverable if the embedding is "unrolled". PCA on this data picks the largest variance direction in $\mathbb{R}^3$, which is the height axis; the rolled $(\cos t, \sin t)$ structure is invisible to PCA.

t-SNE (van der Maaten and Hinton, 2008) addresses the nonlinear case by matching pairwise similarity distributions between high and low dimensions, rather than matching point coordinates by a linear map. The construction uses two probability distributions on the unordered pairs $\{i, j\}$: one in the original space (built from per-point Gaussians, §4.2), one in the embedding (built from a Student-t kernel, §4.4). The objective is the KL divergence between them.

4.2 High-dimensional similarities and perplexity

t-SNE represents high-dimensional similarity by, for each point x_i, a probability distribution over its potential neighbors.

Definition 7.56 (Conditional similarity)**.** For data $x_1, \dots, x_n \in \mathbb{R}^p$ and per-point bandwidths $\sigma_1, \dots, \sigma_n > 0$, the **conditional similarity** of j given i is

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2/(2\sigma_i^2))}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2/(2\sigma_i^2))} \qquad (j \neq i),$$

with $p_{i|i} := 0$. The vector $P_i := (p_{j|i})_{j \neq i}$ is a probability distribution on $\{1, \dots, n\} \setminus \{i\}$.

The bandwidth σ_i controls how concentrated P_i is around the closest neighbors. To choose σ_i, t-SNE matches a user-specified **perplexity**, an effective neighborhood size.

Definition 7.57 (Perplexity)**.** The **perplexity** of P_i is

$$\mathrm{Perp}(P_i) = 2^{H(P_i)}, \qquad H(P_i) = -\sum_{j \neq i} p_{j|i} \log_2 p_{j|i}.$$

Perplexity ranges from 1 (when P_i is concentrated at a single point) to $n-1$ (when P_i is uniform on $\{1,\dots,n\}\setminus\{i\}$).

The user specifies a target perplexity $\mathrm{Perp}^\star \in [5,50]$, and each σ_i is chosen so $\mathrm{Perp}(P_i) = \mathrm{Perp}^\star$.

Lemma 7.58 (Perplexity is monotone in bandwidth)**.** For each i with the squared distances $\{d_{ij}^2 := \|x_i - x_j\|^2\}_{j\neq i}$ not all equal, $H(P_i)$ is strictly increasing in σ_i, and so is $\mathrm{Perp}(P_i)$.

Proof. Reparametrize using $\beta := 1/(2\sigma_i^2) > 0$, so σ_i is decreasing in β. Let $Z(\beta) := \sum_{k\neq i} e^{-\beta d_{ik}^2}$ and write $H(P_i) = H(\beta)$ as a function of β. From $p_{j|i} = e^{-\beta d_{ij}^2}/Z(\beta)$,

$$H(\beta) \;=\; -\sum_j p_{j|i}\log p_{j|i} \;=\; \beta\sum_j p_{j|i}d_{ij}^2 + \log Z(\beta) \;=\; \beta\,\langle d^2\rangle_\beta + \log Z(\beta),$$

where $\langle d^2\rangle_\beta := \sum_j p_{j|i}d_{ij}^2$ is the mean of d^2 under $P_i(\beta)$ (logarithm in any fixed base; the choice affects only a positive constant). The Gibbs identity $d\log Z/d\beta = -\langle d^2\rangle_\beta$ (differentiate $\log Z$ once and recognize the result as a P_i-expectation) gives

$$\frac{dH}{d\beta} = \langle d^2\rangle_\beta + \beta\frac{d\langle d^2\rangle_\beta}{d\beta} - \langle d^2\rangle_\beta = \beta\frac{d\langle d^2\rangle_\beta}{d\beta}.$$

The Gibbs second identity $d\langle d^2\rangle_\beta/d\beta = -\mathrm{Var}_\beta(d^2)$ yields

$$\frac{dH}{d\beta} = -\beta\,\mathrm{Var}_\beta(d^2) < 0,$$

strict whenever the squared distances are not all equal under $P_i(\beta)$. H is therefore strictly decreasing in β, equivalently strictly increasing in σ_i; $\mathrm{Perp}(P_i) = 2^{H(P_i)}$ inherits the same monotonicity. $\square$

The lemma justifies binary search on σ_i. Standard implementations (`Rtsne`) bisect to a tolerance of 10^{-5} on $\mathrm{Perp}(P_i) - \mathrm{Perp}^\star$.

4.3 Symmetrized joint distribution

The conditional similarities $p_{j|i}$ are not symmetric in (i, j). The KL objective compares them to a symmetric low-dimensional similarity (§4.4), so we symmetrize p to obtain a single joint distribution.

Definition 7.59 (Symmetrized similarity)**.** The **symmetrized similarity** is

$$p_{ij} := \frac{p_{j|i} + p_{i|j}}{2n} \qquad (i \neq j).$$

p_{ij} is symmetric ($p_{ij} = p_{ji}$); the family $\{p_{ij}\}$ is a probability distribution on the ordered pairs $\{(i, j) : i \neq j\}$.

Lemma 7.60 (Normalization and outlier robustness)**.**

(i) $\sum_{i \neq j} p_{ij} = 1$.

(ii) For every i, $\sum_{j \neq i} p_{ij} \geqslant 1/(2n)$.

Proof. (i) Each $\sum_{j \neq i} p_{j|i} = 1$, summed over i gives n; hence $\sum_{i \neq j} p_{ij} = (1/(2n)) \sum_{i \neq j}(p_{j|i} + p_{i|j}) = (1/(2n))(n + n) = 1$.

(ii) $\sum_{j \neq i} p_{ij} = (1/(2n))(\sum_{j \neq i} p_{j|i} + \sum_{j \neq i} p_{i|j}) = (1/(2n))(1 + \sum_{j \neq i} p_{i|j}) \geqslant 1/(2n)$, since $\sum_{j \neq i} p_{i|j} \geqslant 0$. $\square$

The lower bound (ii) is the outlier-robustness motivation of van der Maaten and Hinton (2008): under the asymmetric formulation, an outlier i with large σ_i has small $p_{j|i}$ for every j, contributing weakly to the objective. The symmetric form ensures every i's row marginal $\sum_j p_{ij} \geqslant 1/(2n)$ regardless of σ_i.

4.4 The crowding problem and the Student-t low-dimensional kernel

t-SNE's low-dimensional kernel is heavy-tailed Student-t with one degree of freedom (Cauchy), not Gaussian. The choice is driven by a geometric problem about the embedding dimension.

Definition 7.61 (Crowding problem)**.** The **crowding problem** is the failure to fit local-distance and moderate-distance structure simultaneously when embedding into low dimensions ($d = 2$ or 3). In high-dimensional data, a point can have many neighbors at moderate distance; in $\mathbb{R}^d$ for small d, the volume of a ball at radius r scales as r^d, which is too small to accommodate them all at appropriate distances under a Gaussian low-dimensional kernel.

Definition 7.62 (Low-dimensional similarity)**.** For embedded points $y_1, \ldots, y_n \in \mathbb{R}^d$, the **low-dimensional similarity** is

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l}(1 + \|y_k - y_l\|^2)^{-1}} \qquad (i \neq j),$$

with $q_{ii} := 0$. The numerator $(1 + \|y_i - y_j\|^2)^{-1}$ is the unnormalized Student-t_1 kernel at squared distance $\|y_i - y_j\|^2$.

The Student-t_1 kernel addresses crowding by allowing moderate-distance pairs to occupy correspondingly larger embedding distances without losing similarity weight as fast as Gaussian.

Lemma 7.63 (Heavy tails)**.** The Gaussian kernel $\exp(-r^2/2)$ decays super-polynomially in r^2: for every $k \geqslant 1$, $\exp(-r^2/2) = o(r^{-2k})$ as $r \to \infty$. The Student-t_1 kernel $(1 + r^2)^{-1}$ decays polynomially: $(1 + r^2)^{-1} \sim r^{-2}$. The Student-$t_1$ tail is therefore heavier than the Gaussian tail.

Proof. Both rates follow from elementary calculus. The polynomial-vs-exponential comparison $\lim_{r\to\infty} r^{2k} \exp(-r^2/2) = 0$ holds for every k by L'Hôpital, giving $\exp(-r^2/2) = o(r^{-2k})$. Direct expansion gives $(1+r^2)^{-1} = r^{-2}(1+r^{-2})^{-1} \to r^{-2}$ as $r \to \infty$. Hence $(1+r^2)^{-1}/\exp(-r^2/2) \to \infty$. □

The optimization consequence: a high-dim point x_j at moderate distance from x_i with moderate p_{ij} can be embedded at correspondingly moderate $\|y_i - y_j\|$ in $\mathbb{R}^d$, and the Student-t_1 kernel still produces a comparable q_{ij}. Under a Gaussian low-dim kernel (as in the predecessor SNE algorithm), matching the same moderate p_{ij} would require a small embedding distance, compressing all moderate-similarity pairs to short distances and producing the crowded plot.

4.5 KL-divergence objective

Definition 7.64 (t-SNE objective)**.** For symmetrized high-dim similarities p_{ij} (Definition 7.59) and low-dim similarities q_{ij} (Definition 7.62), the **t-SNE cost function** is

$$C(\boldsymbol{y}) \;=\; D_{\mathrm{KL}}(P \,\|\, Q) \;=\; \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}, \qquad \boldsymbol{y} \;=\; (y_1, \ldots, y_n) \in \mathbb{R}^{n \times d}.$$

KL is asymmetric in (P, Q): large p_{ij} requires large q_{ij} (heavy penalty for separating in the embedding what was close in high-dim), while small p_{ij} does not require small q_{ij} (mild penalty for placing close in the embedding what was far in high-dim). Local structure (high-p pairs) is therefore preserved more carefully than global structure (low-p pairs).

4.6 Gradient derivation

Theorem 7.65 (t-SNE gradient)**.** With $C(\boldsymbol{y})$ as in Definition 7.64 and $\boldsymbol{y} \in \mathbb{R}^{n\times d}$,

$$\frac{\partial C}{\partial y_a} = 4\sum_{j\neq a}(p_{aj} - q_{aj})\, q_{aj}\, Z\,(y_a - y_j),$$

where $Z := \sum_{k\neq l}(1 + \|y_k - y_l\|^2)^{-1}$ is the normalizer of q. Equivalently, using $q_{aj}Z = (1 + \|y_a - y_j\|^2)^{-1}$,

$$\frac{\partial C}{\partial y_a} = 4\sum_{j\neq a}(p_{aj} - q_{aj})\,(1 + \|y_a - y_j\|^2)^{-1}\,(y_a - y_j).$$

Proof of Theorem 7.65

Set $r_{ij} := (1 + \|y_i - y_j\|^2)^{-1}$ for $i \neq j$, so $q_{ij} = r_{ij}/Z$. Decompose

$$C = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log r_{ij} + \log Z \cdot \sum_{i \neq j} p_{ij}.$$

The first term is constant in $\boldsymbol{y}$; the third reduces to $\log Z$ since $\sum_{i \neq j} p_{ij} = 1$ (Lemma 7.60). Hence

$$\frac{\partial C}{\partial y_a} = -\sum_{i \neq j} p_{ij} \frac{\partial \log r_{ij}}{\partial y_a} + \frac{\partial \log Z}{\partial y_a}.$$

Per-pair derivative. For $r_{ij} = (1 + \|y_i - y_j\|^2)^{-1}$, write $s_{ij} := 1 + \|y_i - y_j\|^2$. Then $\partial s_{ij}/\partial y_i = 2(y_i - y_j)$ and $\log r_{ij} = -\log s_{ij}$, so

$$\frac{\partial \log r_{ij}}{\partial y_i} = -\frac{1}{s_{ij}} \cdot 2(y_i - y_j) = -2r_{ij}(y_i - y_j).$$

By symmetry of s_{ij} in (i, j), $\partial \log r_{ij}/\partial y_j = +2r_{ij}(y_i - y_j)$. For $a \notin \{i, j\}$, the derivative is zero.

First sum. Contributions to $\sum_{i\neq j} p_{ij}\partial \log r_{ij}/\partial y_a$ come only from pairs (i,j) with $a \in \{i,j\}$:

$$\begin{aligned}\sum_{i\neq j} p_{ij}\frac{\partial \log r_{ij}}{\partial y_a} &= \sum_{j\neq a} p_{aj}(-2r_{aj})(y_a - y_j) + \sum_{i\neq a} p_{ia}(+2r_{ia})(y_i - y_a)\\ &= -2\sum_{j\neq a} p_{aj}r_{aj}(y_a - y_j) - 2\sum_{i\neq a} p_{ia}r_{ia}(y_a - y_i),\end{aligned}$$

where the second line rewrites $(y_i - y_a) = -(y_a - y_i)$ and absorbs the sign. Using $p_{ij} = p_{ji}$ and $r_{ij} = r_{ji}$ to merge the two sums (the dummy index in the second sum can be relabelled j):

$$\sum_{i\neq j} p_{ij}\frac{\partial \log r_{ij}}{\partial y_a} = -4\sum_{j\neq a} p_{aj}r_{aj}(y_a - y_j).$$

Second sum. $\partial Z/\partial y_a = \sum_{k\neq l} \partial r_{kl}/\partial y_a$. From $\partial r_{ij}/\partial y_i = -2r_{ij}^2(y_i - y_j)$ and the analogous symmetric statement for $\partial/\partial y_j$, the same merging argument as above gives

$$\frac{\partial Z}{\partial y_a} = -4\sum_{j\neq a} r_{aj}^2(y_a - y_j),$$

so $\partial \log Z/\partial y_a = (1/Z)\partial Z/\partial y_a = -4\sum_{j\neq a} q_{aj}r_{aj}(y_a - y_j)$.

Combine.

$$\begin{aligned}\frac{\partial C}{\partial y_a} &= -\left(-4\sum_{j\neq a} p_{aj}r_{aj}(y_a - y_j)\right) + \left(-4\sum_{j\neq a} q_{aj}r_{aj}(y_a - y_j)\right)\\ &= 4\sum_{j\neq a}(p_{aj} - q_{aj})r_{aj}(y_a - y_j)\\ &= 4\sum_{j\neq a}(p_{aj} - q_{aj})q_{aj}Z(y_a - y_j),\end{aligned}$$

using $r_{aj} = q_{aj}Z$. The second display in the theorem statement is the form with r_{aj} kept explicit; the two are equivalent. ■

The gradient has a physical interpretation. Define the per-pair force $F_{aj} := 4(p_{aj} - q_{aj})r_{aj}(y_a - y_j)$. When $p_{aj} > q_{aj}$ (the high-dim pair is closer than the low-dim pair under the current embedding), F_{aj} pulls y_a toward y_j. When $p_{aj} < q_{aj}$ (high-dim is farther than low-dim), F_{aj} pushes y_a away from y_j.

The optimizer adjusts each y_a until the net force vanishes.

4.7 Optimization: early exaggeration, learning rate, momentum

The objective C is non-convex with many local minima. The standard t-SNE optimizer is gradient descent with momentum and a starting trick.

Initialization. Original t-SNE (van der Maaten and Hinton, 2008) uses $y_i^{(0)} \sim \mathcal{N}(0, 10^{-4} I_d)$ independently. Current practice, especially in single-cell genomics (Kobak and Berens, 2019), instead sets $y^{(0)}$ to the first d principal components of X, rescaled to the small-variance range; this initialization is reproducible across seeds and tends to preserve global structure better. `Rtsne` accepts this via the `Y_init` argument.

Momentum update. For learning rate η and momentum coefficient $\alpha(t) \in [0, 1)$,

$$y_i^{(t+1)} = y_i^{(t)} - \eta \frac{\partial C}{\partial y_i^{(t)}} + \alpha(t)(y_i^{(t)} - y_i^{(t-1)}).$$

Standard hyperparameters: $\eta = 200$, $\alpha(t) = 0.5$ for $t \leqslant 250$ and $\alpha(t) = 0.8$ thereafter.

Early exaggeration. For $t \leqslant T_{\text{ee}}$ (typically $T_{\text{ee}} = 250$), replace p_{ij} in the cost by $\alpha_{\text{ee}} \cdot p_{ij}$ for $\alpha_{\text{ee}} > 1$ (typically 4 or 12). The exaggerated p amplifies the attractive forces in the gradient, encouraging clusters to form quickly. After T_{ee}, α_{ee} is set to 1 and the optimizer continues to convergence.

Iterations. A typical run uses 1000 iterations: 250 with early exaggeration, 750 without.

Perplexity. The user-specified target $\text{Perp}^\star$ controls effective neighborhood size, with $[5, 50]$ the standard range. Smaller perplexity emphasizes fine local structure; larger perplexity smooths over local detail and produces more global structure.

Computational cost. Naive computation of all p_{ij} requires $O(n^2)$ pairwise distances; the gradient also requires $O(n^2)$ work per iteration via the Z normalizer. Barnes–Hut t-SNE (van der Maaten, 2014) approximates the long-range gradient terms via a quadtree (in $\mathbb{R}^2$) or octree (in $\mathbb{R}^3$), reducing per-iteration cost to $O(n \log n)$. `Rtsne` uses Barnes–Hut by default.

Newer competitors: TriMap and PaCMAP

Two more recent nonlinear DR methods are sometimes used in place of t-SNE. **TriMap** (Amid and Warmuth, 2019) replaces pairwise comparisons with triplet constraints (each anchor with one near and one far neighbor) to better preserve global structure. **PaCMAP** (Wang et al., 2021) balances near, mid-near, and far pairs explicitly, aiming to preserve global structure while keeping local structure. Neither has displaced t-SNE or UMAP in mainstream R workflows; both are most readily available in Python.

5 UMAP: Uniform Manifold Approximation and Projection

5.1 The Riemannian-manifold assumption

UMAP (McInnes et al., 2018) models the high-dimensional data as samples from a distribution on a Riemannian manifold M embedded in $\mathbb{R}^p$. Two technical assumptions justify the local fuzzy-neighborhood construction of §5.2:

(A1) The data distribution on M is approximately uniform with respect to the manifold's natural volume measure.

(A2) The Riemannian metric on M is locally constant near each x_i: in a small ball around x_i, distances are well approximated by Euclidean distances multiplied by a point-specific factor.

(A1) and (A2) are local idealizations. Their role is to license a per-point Euclidean approximation: in the small-σ_i neighborhood of each x_i, the data look like a Euclidean ball whose radius is calibrated to match the local sample density. The next subsection turns this into a numerical construction.

5.2 Local fuzzy neighborhoods

For each data point x_i, UMAP builds a fuzzy 1-simplex (an edge) to every other data point with a strength of membership in $(0, 1]$ derived from local distances.

Definition 7.66 (Local fuzzy weight)**.** For data $x_1, \ldots, x_n \in \mathbb{R}^p$ with metric d, and a hyperparameter $k \in \{2, \ldots, n-1\}$ (the number-of-neighbors parameter), define for each i:

$$\rho_i := \min_{j \neq i} d(x_i, x_j) \qquad \text{(distance to the nearest neighbor)},$$

$$w_{ij} := \exp\left(-\frac{\max(0,\ d(x_i, x_j) - \rho_i)}{\sigma_i} \right) \qquad (j \neq i),$$

with $\sigma_i > 0$ chosen so that

$$\sum_{j \neq i} w_{ij} = \log_2 k.$$

The subtraction of ρ_i ensures the nearest-neighbor weight is exactly 1, so the closest neighbor is always "fully connected" to x_i. The choice $\sum_j w_{ij} = \log_2 k$ is UMAP's analog of t-SNE's perplexity, with k playing the role of Perp*. Binary search on σ_i is justified by the same Gibbs-style monotonicity argument as Lemma 7.58: $\sum_j w_{ij}$ is strictly monotone in σ_i (larger σ_i produces a less peaked w, increasing the sum).

5.3 Fuzzy union: probabilistic OR

The weights w_{ij} are asymmetric: $w_{ij} \neq w_{ji}$ in general because $\rho_i \neq \rho_j$ and $\sigma_i \neq \sigma_j$. UMAP symmetrizes via the probabilistic-OR (also called the algebraic-sum) t-conorm.

Definition 7.67 (Fuzzy union of edge memberships)**.** The **symmetric fuzzy edge weight** is

$$\mu_{ij} \;:=\; w_{ij} + w_{ji} - w_{ij}w_{ji} \qquad (i \neq j).$$

Lemma 7.68 (Properties of the fuzzy union)**.**

(i) $\mu_{ij} = \mu_{ji}$ (symmetry).

(ii) $\mu_{ij} \in [0,1]$ when $w_{ij}, w_{ji} \in [0,1]$ (well-defined fuzzy membership).

(iii) $\mu_{ij} = 1 - (1 - w_{ij})(1 - w_{ji})$ (probabilistic-OR identity).

Proof. (i) is immediate from the symmetric definition. (iii) Expanding the right side: $1 - (1 - w_{ij})(1 - w_{ji}) = 1 - 1 + w_{ij} + w_{ji} - w_{ij}w_{ji} = w_{ij} + w_{ji} - w_{ij}w_{ji} = \mu_{ij}$. (ii) From (iii), $\mu_{ij} = 1 - (1 - w_{ij})(1 - w_{ji})$ with both $(1 - w_{ij}), (1 - w_{ji}) \in [0, 1]$, so the product is in $[0, 1]$ and $\mu_{ij} \in [0, 1]$. □

The probabilistic-OR identity is read: under the (independence) assumption that "x_i and x_j are fuzzy-neighbors" from i's perspective is independent of the same from j's perspective, the probability that they are neighbors from at least one perspective is $1-(1-w_{ij})(1-w_{ji})$. This is one of several t-conorms (others include $\max(w_{ij}, w_{ji})$ and $\min(1, w_{ij}+w_{ji})$); UMAP's choice is dictated by the Spivak adjunction, where the algebraic sum is the canonical fuzzy-union under the categorical formalism.

5.4 Spivak's metric realization adjunction

The categorical foundation of UMAP is an adjunction between the categories **FinEPMet** (finite extended-pseudo-metric spaces, Definition 7.35) and **sFuzz** (fuzzy simplicial sets, Definition 7.33). Both were introduced in §1.8.

Definition 7.69 (Standard metric simplex)**.** For $n \geqslant 0$ and $a \in (0,1]$, the **standard metric n-simplex of strength** a is the finite extended-pseudo-metric space

$$\Delta^n_a \;:=\; \big(\{0,1,\dots,n\},\ d_a\big), \qquad d_a(i,j) \;:=\; \begin{cases} 0 & i=j, \\ -\log a & i \neq j. \end{cases}$$

A morphism $\Delta^n_a \to (Y, d_Y)$ in **FinEPMet** is a function $f : \{0,\dots,n\} \to Y$ short with respect to the metrics, equivalently $d_Y(f(i), f(j)) \leqslant -\log a$ for every i,j. Hence

$$\hom_{\mathbf{FinEPMet}}(\Delta^n_a, Y) \;\cong\; \big\{(y_0,\dots,y_n) \in Y^{n+1} : d_Y(y_i, y_j) \leqslant -\log a \text{ for all } i,j\big\}.$$

Definition 7.70 (Singular fuzzy simplicial set)**.** The **singular** functor $\mathrm{Sing} : \mathbf{FinEPMet} \to \mathbf{sFuzz}$ is defined on objects by

$$\mathrm{Sing}(Y)([n], a) \;:=\; \hom_{\mathbf{FinEPMet}}(\Delta^n_a, Y),$$

with face/degeneracy maps in the Δ-direction induced by precomposition with the corresponding morphisms of Δ-objects, and strength-restriction maps in the $\mathcal{I}^{\mathrm{op}}$-direction induced by the inclusions

$$\hom(\Delta^n_b, Y) \hookrightarrow \hom(\Delta^n_a, Y) \qquad (a \leqslant b)$$

that hold because $-\log a \geqslant -\log b$ relaxes the metric constraint. On a morphism $\phi : Y \to Y'$ in **FinEPMet**, $\mathrm{Sing}(\phi)([n], a)$ post-composes with ϕ.

Definition 7.71 (Metric realization)**.** The **metric realization** functor $\mathrm{Real} : \mathbf{sFuzz} \to \mathbf{FinEPMet}$ sends a fuzzy simplicial set X to the colimit

$$\mathrm{Real}(X) := \operatorname{colim}_{\Delta^n_a \to X} \Delta^n_a,$$

taken in **FinEPMet** over the category of elements of X (objects: pairs $((n, a), \sigma)$ with $\sigma \in X([n], a)$; morphisms: Δ- and $\mathcal{I}^{\mathrm{op}}$-morphisms compatible with the σ via the functoriality of X). For a morphism $\eta : X \to X'$ of fuzzy simplicial sets, $\mathrm{Real}(\eta)$ is the induced map of colimits.

For finite X (only finitely many non-degenerate simplices across all (n, a)), the colimit is finite: it is the quotient of $\bigsqcup_{\sigma \in X([n],a)} \Delta^n_a$ by the equivalence relation generated by face / degeneracy / strength-restriction identifications, equipped with the induced shortest-path extended pseudo-metric.

Theorem 7.72 (Spivak's adjunction)**.** (Spivak, 2009). The functor pair Real $\dashv$ Sing is an adjoint pair: there is a bijection

$$\Phi_{X,Y} : \hom_{\mathbf{FinEPMet}}(\mathrm{Real}(X),\, Y) \xrightarrow{\cong} \hom_{\mathbf{sFuzz}}(X,\, \mathrm{Sing}(Y))$$

natural in $X \in \mathbf{sFuzz}$ and $Y \in \mathbf{FinEPMet}$.

Proof of Theorem 7.72

The proof uses the universal property of the colimit defining $\mathrm{Real}(X)$ together with the explicit description of $\mathrm{Sing}(Y)$.

Forward direction. Take $g \in \hom_{\mathbf{FinEPMet}}(\mathrm{Real}(X), Y)$, a metric map. The colimit $\mathrm{Real}(X)$ is equipped with structure morphisms $\iota_\sigma : \Delta^n_a \to \mathrm{Real}(X)$ for each $\sigma \in X([n], a)$. The composition $g \circ \iota_\sigma : \Delta^n_a \to Y$ is a metric map, hence an element of $\mathrm{Sing}(Y)([n], a)$. Define

$$\Phi_{X,Y}(g)_{[n],a}(\sigma) \;:=\; g \circ \iota_\sigma \;\in\; \mathrm{Sing}(Y)([n], a).$$

We verify $\Phi_{X,Y}(g)$ is a natural transformation $X \Rightarrow \mathrm{Sing}(Y)$. For a morphism $\phi : ([m], b) \to$

$([n], a)$ in $\Delta \times \mathcal{I}^{\mathrm{op}}$ and $\sigma \in X([n], a)$, naturality requires

$$\mathrm{Sing}(Y)(\phi)\big(g \circ \iota_\sigma\big) \;=\; g \circ \iota_{X(\phi)(\sigma)}.$$

The colimit defining $\mathrm{Real}(X)$ identifies $\iota_{X(\phi)(\sigma)}$ with ι_σ pre-composed with the $\Delta \times \mathcal{I}^{\mathrm{op}}$-induced morphism $\Delta^m_b \to \Delta^n_a$, exactly the identification that makes the displayed equality hold.

Reverse direction. Take a natural transformation $\eta : X \to \mathrm{Sing}(Y)$. Each component $\eta_{[n],a}(\sigma) \in \mathrm{Sing}(Y)([n], a) = \hom(\Delta^n_a, Y)$ is a metric map $f_\sigma : \Delta^n_a \to Y$. Naturality of η asserts that for every morphism $\phi : ([m], b) \to ([n], a)$ in $\Delta \times \mathcal{I}^{\mathrm{op}}$ and every $\sigma \in X([n], a)$, $f_{X(\phi)(\sigma)}$ equals f_σ pre-composed with the corresponding $\Delta^m_b \to \Delta^n_a$. This is precisely the compatibility condition required by the universal property of the colimit defining $\mathrm{Real}(X)$. Hence the family $\{f_\sigma\}$ extends uniquely to a metric map

$$\Psi_{X,Y}(\eta) \;:=\; \mathrm{colim}\,\{f_\sigma\} \;:\; \mathrm{Real}(X) \to Y,$$

with $\Psi_{X,Y}(\eta) \circ \iota_\sigma = f_\sigma$ for every σ.

Mutual inverses. For $g \in \hom(\mathrm{Real}(X), Y)$, $\Psi(\Phi(g))$ is the unique metric map satisfying $\Psi(\Phi(g)) \circ \iota_\sigma = g \circ \iota_\sigma$ for every σ. By the universal property's uniqueness clause, $\Psi(\Phi(g)) = g$.

For $\eta : X \to \mathrm{Sing}(Y)$, $\Phi(\Psi(\eta))_{[n],a}(\sigma) = \Psi(\eta) \circ \iota_\sigma = f_\sigma = \eta_{[n],a}(\sigma)$ by construction. Hence $\Phi(\Psi(\eta)) = \eta$.

Naturality. For $f : Y \to Y'$ in **FinEPMet** and $g \in \hom(\mathrm{Real}(X), Y)$,

$$\Phi_{X,Y'}(f \circ g)_{[n],a}(\sigma) \;=\; f \circ g \circ \iota_\sigma \;=\; \mathrm{Sing}(f)\big(\Phi_{X,Y}(g)_{[n],a}(\sigma)\big),$$

so Φ is natural in Y. For $\eta : X' \to X$ in **sFuzz** and $g \in \hom(\mathrm{Real}(X), Y)$, $\mathrm{Real}(\eta) : \mathrm{Real}(X') \to \mathrm{Real}(X)$ satisfies $\mathrm{Real}(\eta) \circ \iota_{\sigma'} = \iota_{\eta(\sigma')}$ for $\sigma' \in X'([n], a)$, and

$$\Phi_{X',Y}(g \circ \mathrm{Real}(\eta))_{[n],a}(\sigma') \;=\; g \circ \mathrm{Real}(\eta) \circ \iota_{\sigma'} \;=\; g \circ \iota_{\eta(\sigma')} \;=\; \Phi_{X,Y}(g)_{[n],a}(\eta(\sigma')),$$

so Φ is natural in X via $\mathrm{Sing}(Y)$ post-composition with η. ■

Example 7.6 (Unit and counit on a single edge)**.** By Theorem 7.28, the adjunction $\mathrm{Real} \dashv \mathrm{Sing}$ has a unit $\eta : 1_{\mathbf{sFuzz}} \Rightarrow \mathrm{Sing} \circ \mathrm{Real}$ and counit $\varepsilon : \mathrm{Real} \circ \mathrm{Sing} \Rightarrow 1_{\mathbf{FinEPMet}}$. Concretely:

- $\eta_X : X \to \mathrm{Sing}(\mathrm{Real}(X))$ sends $\sigma \in X([n], a)$ to the metric map $\iota_\sigma : \Delta^n_a \to \mathrm{Real}(X)$, viewed as an element of $\mathrm{Sing}(\mathrm{Real}(X))([n], a)$.

- $\varepsilon_Y : \mathrm{Real}(\mathrm{Sing}(Y)) \to Y$ is the unique metric map whose composition with ι_f (for $f \in \mathrm{Sing}(Y)([n], a)$) recovers $f : \Delta^n_a \to Y$ itself.

The triangle identities of Theorem 7.28 are then a formal consequence of the bijection in Theorem 7.72.

5.5 The functorial reconstruction principle

The Spivak adjunction underwrites UMAP's optimization target. The high-dim data, treated as an extended-pseudo-metric space $(\{x_1, \ldots, x_n\}, d)$, is canonically associated with a fuzzy simplicial set $\mathrm{Sing}((\{x_i\}, d)) \in \mathbf{sFuzz}$ via the right adjoint. A candidate low-dim embedding $(y_1, \ldots, y_n) \in \mathbb{R}^d$ is similarly associated with $\mathrm{Sing}((\{y_i\}, d_\Phi))$ for a low-dim metric d_Φ derived from the kernel Φ of §5.6. The optimization minimizes a divergence between these two fuzzy simplicial sets.

In practice, UMAP works only with the 1-skeleton: the $n = 1$ part of each Sing. The fuzzy 1-simplices (edges) carry the membership weights μ_{ij} (§5.3); higher-dimensional simplices, while present in principle, do not appear in the algorithmic objective.

1-skeleton in practice, full simplicial structure in theory

The full categorical machinery licenses the construction by exhibiting it as the canonical adjunction-induced object on the metric space. Once the 1-skeleton is in hand, optimization proceeds on edges only, and the higher-dimensional structure plays no algorithmic role. The simplicial-set framework is therefore a justification of the construction's well-definedness rather than a description of the algorithm's data flow.

5.6 Low-dimensional Cauchy-like kernel

UMAP's low-dimensional similarity is a Cauchy-like kernel parameterized by two scalars.

Definition 7.73 (Low-dim membership)**.** For embedded points $y_1, \ldots, y_n \in \mathbb{R}^d$ and parameters $a > 0$, $b > 0$, the **low-dim edge membership** is

$$\nu_{ij} \;:=\; \Phi(y_i, y_j) \;:=\; \frac{1}{1 + a\|y_i - y_j\|^{2b}} \qquad (i \neq j).$$

The parameters a, b are chosen by least-squares fitting of Φ to a target piecewise function depending on a hyperparameter min_dist:

$$\psi(d) = \begin{cases} 1 & d \leqslant \text{min\_dist}, \\ \exp\big(-(d - \text{min\_dist})\big) & d > \text{min\_dist}. \end{cases}$$

Solving $\min_{a,b} \int_0^\infty (\Phi(d) - \psi(d))^2 \, dd$ for typical min_dist $\in [0.001, 0.5]$ gives (a, b) tabulated in McInnes–Healy (e.g., min_dist $= 0.1$ yields $(a, b) \approx (1.577, 0.895)$). The Cauchy-like form $(1 + a\|\cdot\|^{2b})^{-1}$ generalizes the t-SNE Student-t_1 kernel ($a = b = 1$) and inherits the heavy-tail behavior addressing crowding (§4.4).

5.7 Cross-entropy objective and gradient

Definition 7.74 (UMAP objective)**.** The **UMAP cost function** is the fuzzy cross-entropy of edge memberships (Definition 7.18) applied to μ_{ij} and ν_{ij}:

$$C(\boldsymbol{y}) = \sum_{i<j} \left[\mu_{ij} \log \frac{\mu_{ij}}{\nu_{ij}} + (1 - \mu_{ij}) \log \frac{1 - \mu_{ij}}{1 - \nu_{ij}} \right],$$

where the sum is over unordered pairs $\{i, j\}$ with $\mu_{ij}, \nu_{ij} \in (0, 1)$ to avoid logarithm boundary cases.

The high-dim part μ_{ij} is fixed (computed from the data), so C depends on $\boldsymbol{y}$ only through ν_{ij}.

Theorem 7.75 (UMAP gradient)**.**

$$\frac{\partial C}{\partial y_i} = 2ab \sum_{j \neq i} \left[\mu_{ij} - \frac{(1-\mu_{ij})\,\nu_{ij}}{1-\nu_{ij}} \right] \nu_{ij} \, \|y_i - y_j\|^{2b-2} \, (y_i - y_j).$$

Proof of Theorem 7.75

Set $r_{ij} := \|y_i - y_j\|^2$ for $i \neq j$, so $\nu_{ij} = (1 + a r_{ij}^b)^{-1}$. The cost decomposes as

$$C = \sum_{i<j} \left[\mu_{ij} \log \mu_{ij} - \mu_{ij} \log \nu_{ij} + (1-\mu_{ij}) \log(1-\mu_{ij}) - (1-\mu_{ij}) \log(1-\nu_{ij}) \right].$$

The μ-only terms are constant in $\boldsymbol{y}$, giving

$$\frac{\partial C}{\partial y_a} = \sum_{i<j} \left[-\frac{\mu_{ij}}{\nu_{ij}} + \frac{1-\mu_{ij}}{1-\nu_{ij}} \right] \frac{\partial \nu_{ij}}{\partial y_a}.$$

Per-pair derivative. $\nu_{ij} = (1 + a r_{ij}^b)^{-1}$ has

$$\frac{\partial \nu_{ij}}{\partial r_{ij}} = -\nu_{ij}^2 \cdot ab r_{ij}^{b-1}, \qquad \frac{\partial r_{ij}}{\partial y_i} = 2(y_i - y_j),$$

so by the chain rule, for $a \in \{i, j\}$,

$$\frac{\partial \nu_{ij}}{\partial y_a} = -2ab\, \nu_{ij}^2\, r_{ij}^{b-1} \cdot \begin{cases} (y_a - y_j) & a = i, \\ (y_a - y_i) & a = j, \end{cases}$$

and zero otherwise. Both cases collapse to $\partial \nu_{ij} / \partial y_a = -2ab\nu_{ij}^2 r_{ij}^{b-1}(y_a - y_{j'})$ where j' is the other index in $\{i, j\}$.

Sum over pairs. The pairs containing a are (a, j) for $j \neq a$. Each contributes once to the unordered-pair sum (with $i = a < j$ if $a < j$ or $j < i = a$ if $j < a$, equivalently $\{i, j\} = \{a, j\}$). Hence

$$\frac{\partial C}{\partial y_a} = \sum_{j \neq a} \left[-\frac{\mu_{aj}}{\nu_{aj}} + \frac{1 - \mu_{aj}}{1 - \nu_{aj}} \right] \cdot \big(-2ab\big)\, \nu_{aj}^2\, r_{aj}^{b-1}\, (y_a - y_j).$$

Distribute $-2ab\,\nu_{aj}^2$ over the bracket, using $\nu_{aj}^2 \cdot (\mu_{aj}/\nu_{aj}) = \mu_{aj}\nu_{aj}$ and $\nu_{aj}^2 \cdot (1-\mu_{aj})/(1-\nu_{aj}) = (1-\mu_{aj})\nu_{aj}^2/(1-\nu_{aj})$; the overall factor of -1 from outside the bracket flips the order of the two terms:

$$\frac{\partial C}{\partial y_a} = 2ab \sum_{j \neq a} \left[\mu_{aj}\nu_{aj} - \frac{(1-\mu_{aj})\,\nu_{aj}^2}{1-\nu_{aj}} \right] r_{aj}^{b-1}\,(y_a - y_j).$$

Factoring ν_{aj} out of the bracket gives the stated form. ■

The bracket has an attractive / repulsive reading parallel to t-SNE. When μ_{aj} is large and ν_{aj} small (high-dim closer than low-dim), the bracket is positive, $\partial C/\partial y_a$ points along $(y_a - y_j)$, and gradient descent pulls y_a toward y_j. When μ_{aj} is small and ν_{aj} large (high-dim farther than low-dim), the bracket is negative and descent pushes y_a away from y_j.

5.8 Spectral initialization via the graph Laplacian

UMAP initializes $\boldsymbol{y}$ from the bottom non-trivial eigenvectors of the graph Laplacian of the symmetric weight matrix $A_{ij} := \mu_{ij}$ (with $A_{ii} := 0$).

Definition 7.76 (Graph Laplacian)**.** For a symmetric weight matrix $A \in \mathbb{R}^{n\times n}$ with $A_{ij} \geqslant 0$ and $A_{ii} = 0$, the (**unnormalized**) **graph Laplacian** is $L := D - A$, where $D = \operatorname{diag}(\sum_j A_{1j}, \dots, \sum_j A_{nj})$ is the degree matrix. L is symmetric PSD with $L\mathbf{1} = \mathbf{0}$.

The connection to a smoothness functional uses the identity $\sum_{i,j} A_{ij}(z_i - z_j)^2 = 2z^\top Lz$ for any $z \in \mathbb{R}^n$, expanding the square and applying $A_{ij} = A_{ji}$.

Theorem 7.77 (Laplacian eigenmaps)**.** (Belkin and Niyogi, 2003). Let $0 = \lambda_1 \leqslant \lambda_2 \leqslant \cdots \leqslant \lambda_n$ be the eigenvalues of L with orthonormal eigenvectors $u_1, \dots, u_n$ (so $u_1 \propto \mathbf{1}$). The matrix $Y = [u_2 \ \cdots \ u_{d+1}] \in \mathbb{R}^{n\times d}$ minimizes

$$\operatorname{tr}(Y^\top LY) \;=\; \tfrac{1}{2}\sum_{i,j} A_{ij}\|y_i^\top - y_j^\top\|^2$$

subject to $Y^\top Y = I_d$ and $Y^\top\mathbf{1} = \mathbf{0}$. The minimum value is $\sum_{k=2}^{d+1} \lambda_k$.

Proof. Set $W = [u_2 \ \cdots \ u_n] \in \mathbb{R}^{n\times(n-1)}$ orthonormal columns, spanning $\mathbf{1}^\perp$. Any Y with $Y^\top\mathbf{1} = \mathbf{0}$ and $Y^\top Y = I_d$ can be written $Y = WM$ for some $M \in \mathbb{R}^{(n-1)\times d}$ with $M^\top M = I_d$. Then $LW = W\operatorname{diag}(\lambda_2, \ldots, \lambda_n)$ since each u_k for $k \geqslant 2$ is an eigenvector, so

$$Y^\top LY = M^\top W^\top LWM = M^\top \operatorname{diag}(\lambda_2, \ldots, \lambda_n)M.$$

The trace $\operatorname{tr}(M^\top \operatorname{diag}(\lambda_2, \ldots, \lambda_n)M) = \sum_{k=2}^n \lambda_k \|M_{(k-1),:}\|^2$, where $M_{(k-1),:}$ is the $(k-1)$-th row of M. The constraint $M^\top M = I_d$ implies $\sum_{k=2}^n \|M_{(k-1),:}\|^2 = d$ and $\|M_{(k-1),:}\|^2 \in [0, 1]$ for each k. Minimizing $\sum_{k\geqslant 2} \lambda_k \|M_{(k-1),:}\|^2$ over this constraint set is achieved by placing mass 1 on the smallest λ_k, namely $k = 2, \ldots, d+1$, giving M such that WM takes columns $u_2, \ldots, u_{d+1}$. The minimum value is $\sum_{k=2}^{d+1} \lambda_k$. □

The bottom non-trivial eigenvectors of L minimize the smoothness penalty $\sum_{ij} A_{ij}\|y_i - y_j\|^2$, hence preserve the connectivity structure encoded in A. UMAP uses this initialization as a starting point that already separates clusters by graph topology, before the cross-entropy objective refines it locally.

5.9 Stochastic gradient descent with negative sampling

The full UMAP gradient (Theorem 7.75) requires summing over all $\binom{n}{2}$ pairs per epoch, prohibitive for n in the tens of thousands. UMAP instead uses stochastic gradient descent with negative sampling, in the spirit of word2vec.

Per-step update. Each step samples one positive edge $\{i, j\}$ with $\mu_{ij} > 0$ (proportional to μ_{ij}) and a small number m of negative edges $\{i, k\}$ with k sampled uniformly. The attractive gradient at the positive edge ($\partial C/\partial y_i$ with the bracket $-\mu_{aj}$ dominant when μ_{ij} is large) and the repulsive gradient at the negative edges ($\partial C/\partial y_i$ with the bracket $(1 - \mu_{ak})\nu_{ak}/(1 - \nu_{ak})$ dominant when μ_{ik} is small) are applied to update y_i and y_j (or y_k).

Hyperparameters. The number of negative samples per positive sample m is typically 5. The number of epochs is 200 for small n, scaling down to 30 for large n. The learning rate decreases linearly from an initial value (typically 1.0) to 0 over the run.

Computational cost. Per epoch, the algorithm samples $O(|\text{edges}|)$ positives, where $|\text{edges}| \leqslant nk$ (only k neighbors per point have $\mu_{ij} > 0$). Total cost per epoch is $O(nkmd)$, linear in n. The implementation in `uwot` (R) uses approximate nearest-neighbor search (Annoy or HNSW) for the k-NN computation; the empirical end-to-end scaling is $O(n^{1.14})$.

6 Implementation Tour

6.1 PCA via prcomp and step_pca

PCA is a one-line call in base R; the rotation matrix and scores are accessed via standard slots.

```r
data(iris); X <- as.matrix(iris[, 1:4])
pc <- prcomp(X, center = TRUE, scale. = TRUE)
summary(pc)$importance[, 1:3]              # variance explained per PC
```

```
##                             PC1       PC2       PC3
## Standard deviation     1.708361 0.9560494 0.3830886
## Proportion of Variance 0.729620 0.2285100 0.0366900
## Cumulative Proportion  0.729620 0.9581300 0.9948200
```

```r
round(pc$rotation, 3)                      # loadings (right singular vectors)
```

```
##                 PC1    PC2    PC3    PC4
## Sepal.Length  0.521 -0.377  0.720  0.261
## Sepal.Width  -0.269 -0.923 -0.244 -0.124
## Petal.Length  0.580 -0.024 -0.142 -0.801
## Petal.Width   0.565 -0.067 -0.634  0.524
```

```r
head(pc$x, 3)                                  # principal scores
```

```
##            PC1        PC2        PC3        PC4
## [1,] -2.257141 -0.4784238  0.1272796 0.02408751
## [2,] -2.074013  0.6718827  0.2338255 0.10266284
## [3,] -2.356335  0.3407664 -0.0440539 0.02828231
```

In a tidymodels pipeline, PCA is a preprocessing step inserted via `recipes::step_pca`; the number of components k is part of the workflow and tunable.

```r
library(tidymodels)
rec <- recipe(Species ~ ., data = iris) %>%
  step_normalize(all_numeric_predictors()) %>%
  step_pca(all_numeric_predictors(), num_comp = 2)
prep(rec) %>% bake(new_data = NULL) %>% head(3)
```

```
## # A tibble: 3 x 3
##   Species   PC1    PC2
##   <fct>   <dbl>  <dbl>
## 1 setosa  -2.26 -0.478
## 2 setosa  -2.07  0.672
```

```
## 3 setosa  -2.36  0.341
```

6.2 K-means and hierarchical clustering via base R

`kmeans()` runs Lloyd's algorithm (Definition 7.46) with multiple restarts via `nstart`. `hclust()` builds the dendrogram from a precomputed distance matrix.

```r
set.seed(7)
km <- kmeans(scale(X), centers = 3, nstart = 25)
km$tot.withinss                        # within-cluster sum of squares
```

```
## [1] 138.8884
```

```r
table(true = iris$Species, km = km$cluster)
```

```
##             km
## true          1  2  3
##   setosa      0  0 50
##   versicolor 39 11  0
##   virginica  14 36  0
```

```r
d  <- dist(scale(X))
hc <- hclust(d, method = "ward.D2")
```

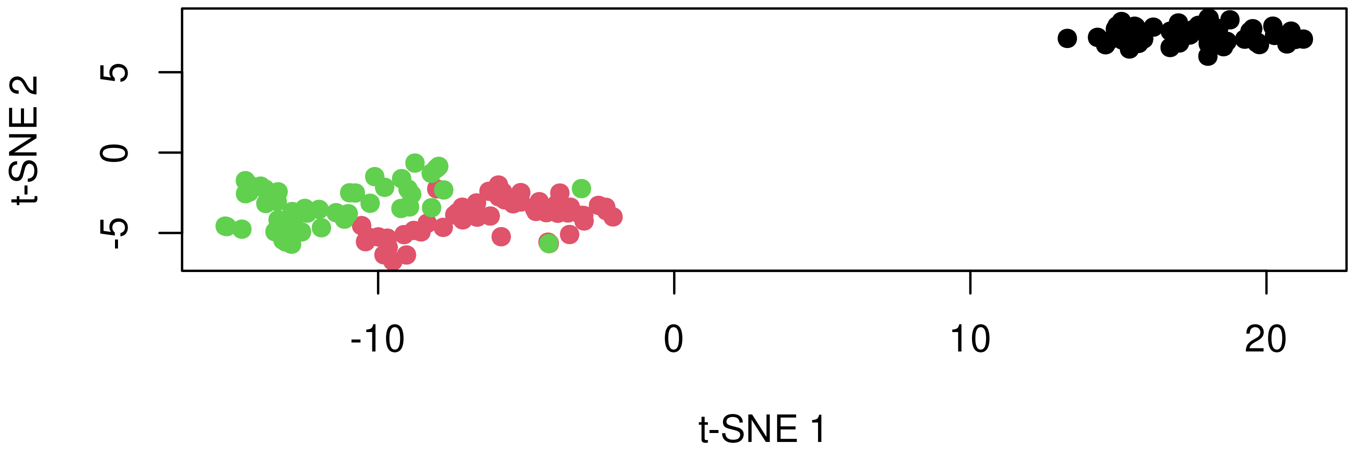
```r
cl <- cutree(hc, k = 3)
table(true = iris$Species, hc = cl)
```

```
##               hc
## true           1  2  3
##   setosa      49  1  0
##   versicolor   0 27 23
##   virginica    0  2 48
```

6.3 t-SNE via Rtsne

`Rtsne::Rtsne` accepts a numeric matrix and returns the embedding in `$Y`. Default `theta = 0.5` enables Barnes–Hut approximation; `theta = 0` requests the exact $O(n^2)$ gradient.

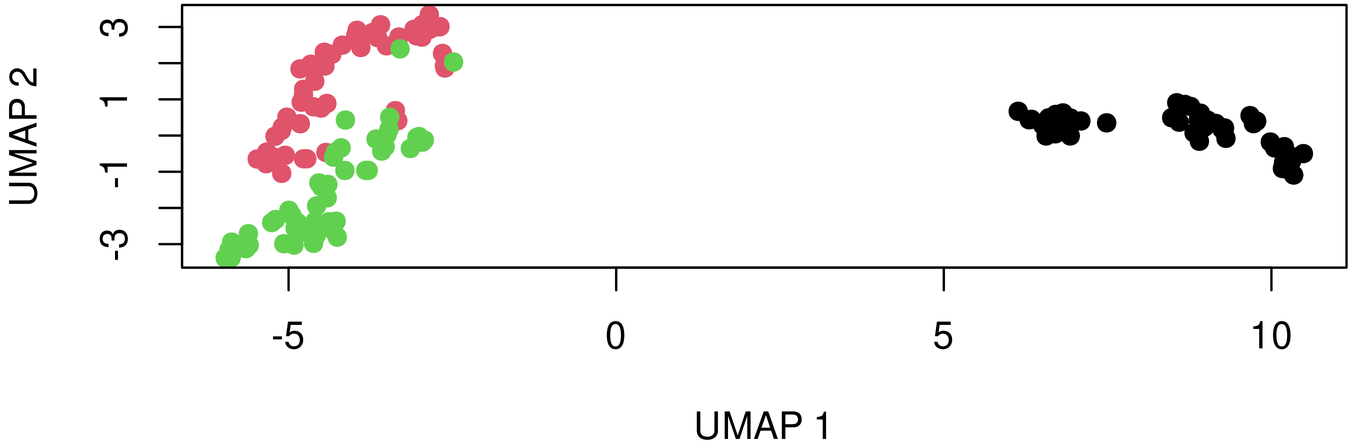
```r
library(Rtsne)
set.seed(7)
em <- Rtsne(scale(X), dims = 2, perplexity = 30, theta = 0.5,
            max_iter = 1000, check_duplicates = FALSE)
plot(em$Y, col = iris$Species, pch = 19,
     xlab = "t-SNE 1", ylab = "t-SNE 2")
```



6.4 UMAP via `uwot`

`uwot::umap` takes the same numeric matrix; `init = "spectral"` (the default) uses Laplacian-eigenmap initialization (§5.8), `init = "random"` switches to the t-SNE-style Gaussian initialization.

```r
library(uwot)
set.seed(7)
em <- umap(scale(X), n_neighbors = 15, min_dist = 0.1,
           n_components = 2, init = "spectral", n_epochs = 200)
plot(em, col = iris$Species, pch = 19,
     xlab = "UMAP 1", ylab = "UMAP 2")
```



6.5 From-scratch t-SNE

The full t-SNE algorithm of §4.6 fits in roughly 20 lines of base R: per-row binary search for bandwidths via `uniroot`, a gradient-descent loop using the matrix form of Theorem 7.65, no momentum or early exaggeration. Output quality on small data ($n \leqslant 500$) is comparable to `Rtsne` but slower.

```r
tsne_min <- function(X, k = 2, perp = 30, n_iter = 500, lr = 200) {
  n  <- nrow(X)
  if (perp >= n - 1) stop("perp must be strictly less than n - 1")
  D2 <- as.matrix(dist(X))^2
  P  <- t(sapply(seq_len(n), function(i) {
    d2 <- D2[i, -i]                          # off-diagonal squared distances
    f <- function(beta) {                    # entropy minus target log_2(perp)
      e <- exp(-beta * (d2 - min(d2)))       # shift to avoid exp underflow
      e <- e / sum(e)
      -sum(e[e > 0] * log2(e[e > 0])) - log2(perp)
    }
    beta <- uniroot(f, c(1e-10, 1e10), tol = 1e-5)$root
    p_i  <- numeric(n)
    p_i[-i] <- exp(-beta * (d2 - min(d2)))
    p_i / sum(p_i)
  }))
  P <- (P + t(P)) / (2 * n); P <- pmax(P, 1e-12)
  Y <- matrix(rnorm(n * k, sd = 1e-4), n, k)
  for (iter in seq_len(n_iter)) {
    DY <- as.matrix(dist(Y))^2
```

```r
    R  <- 1 / (1 + DY); diag(R) <- 0; Q <- R / sum(R)
    PQ <- (P - Q) * R
    grad <- 4 * (Y * rowSums(PQ) - PQ %*% Y)   # vector recycling: Y[i,] * rowSums(PQ)[i]
    Y    <- Y - lr * grad
  }
  Y
}
set.seed(7)
em_scratch <- tsne_min(scale(X), perp = 30, n_iter = 300)
dim(em_scratch); range(em_scratch)
```

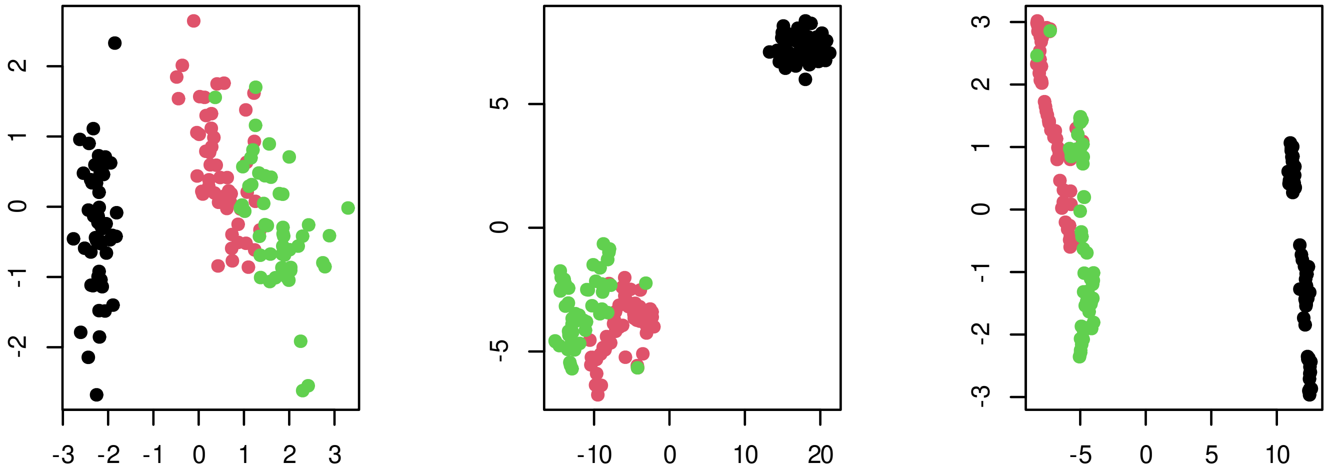
```
## [1] 150   2
## [1] -8.258285 11.665302
```

Mathematically the gradient is $4(D - A)Y$ with $A = (P - Q) \cdot R$ and $D = \mathrm{diag}(\mathrm{rowSums}(A))$, the rewrite of $\sum_j A_{ij}(y_i - y_j) = (\sum_j A_{ij})y_i - \sum_j A_{ij}y_j$ (Theorem 7.65). The R implementation avoids forming D explicitly: `Y * rowSums(A)` computes DY via vector recycling in $O(nk)$ time and constant auxiliary memory, replacing the naive $O(n^2k)$ `diag(rowSums(A))` construction plus matrix multiplication.

6.6 Side-by-side embedding comparison

Putting the four embeddings on the same dataset highlights what each method preserves.

```
library(Rtsne); library(uwot)
set.seed(7); Xs <- scale(X)
em_pca  <- prcomp(Xs)$x[, 1:2]
em_tsne <- Rtsne(Xs, perplexity = 30, check_duplicates = FALSE)$Y
em_umap <- umap(Xs, n_neighbors = 15)
par(mfrow = c(1, 3))
for (em in list(em_pca, em_tsne, em_umap))
  plot(em, col = iris$Species, pch = 19, xlab = "", ylab = "")
```



PCA preserves linear structure and pairwise Euclidean distances on the rank-2 plane; t-SNE pulls Setosa away from the other two species but distorts inter-cluster distances; UMAP separates all three with closer-to-honest global structure (though still parameter-sensitive, §7.10).

7 Pitfalls and Misapplications

7.1 Pitfall 1: PCA without standardization

PCA on raw data with disparately-scaled features makes PC1 align with the highest-variance feature regardless of structure among the others. §2.7 gave the algebraic reason; the practical failure is dramatic.

Tempting because: `prcomp(x)` runs without error or warning on any numeric matrix.

```
library(tidyverse)
set.seed(7); n <- 200
big_scale <- rnorm(n, sd = 100)
small_a   <- rnorm(n, sd = 1)
small_b   <- small_a + rnorm(n, sd = 0.3)
small_c   <- small_a - rnorm(n, sd = 0.3)
X <- cbind(big_scale, small_a, small_b, small_c)
pc_raw <- prcomp(X, scale. = FALSE)
round(pc_raw$rotation[, 1:2], 3)        # PC1 loads almost entirely on big_scale
```

```
##              PC1    PC2
## big_scale -1.000 -0.001
```

```
## small_a   -0.001  0.566
## small_b    0.000  0.577
## small_c    0.000  0.589
```

```r
round(pc_raw$sdev^2 / sum(pc_raw$sdev^2), 3)
```

```
## [1] 1 0 0 0
```

Diagnostic. If PC1's loading is concentrated on a single feature ($|v_{m,1}| \gtrsim 0.95$), suspect a scale issue.

Fix. `prcomp(X, scale. = TRUE)` or pre-standardize via `scale(X)`.

7.2 Pitfall 2: PCA on outlier-laden data

A single distant observation can dominate the sample covariance, making PC1 align toward that observation and hiding structure in the rest.

Tempting because: the sample covariance is sensitive to extreme values, but raw data may not have an obviously flagged outlier; the score plot is accepted at face value.

```r
set.seed(7); n <- 200
X_clean <- matrix(rnorm(n * 4), n, 4)
X <- rbind(X_clean, c(50, 50, 0, 0))      # one extreme row appended
```

```r
pc <- prcomp(scale(X))
range(pc$x[, 1])                          # outlier dominates PC1 score range
```

```
## [1] -19.1820367   0.9410953
```

```r
sort(abs(pc$x[, 1]), decreasing = TRUE)[1:3]
```

```
## [1] 19.1820367  0.9410953  0.8069971
```

Diagnostic. `quantile(pc$x[, 1])` or a score plot reveals one or two observations with extreme PC1 scores.

Fix. Robust PCA via `rrcov::PcaHubert` or `rrcov::PcaCov`; or remove flagged outliers before PCA after diagnosis.

7.3 Pitfall 3: K-means initialization sensitivity

`kmeans()` with `nstart = 1` (R default) returns different cluster assignments on different runs.

Tempting because: the default value is `nstart = 1`; users do not notice unless they re-run.

```r
set.seed(7); n <- 100; K <- 4
centers <- matrix(c(0, 0,  4, 0,  0, 4,  4, 4), ncol = 2, byrow = TRUE)
X <- centers[sample(K, n, replace = TRUE), ] +
```

```
        matrix(rnorm(n * 2, sd = 0.7), n, 2)
sapply(1:5, function(s) {                    # five seeds, nstart = 1 each
  set.seed(s)
  kmeans(X, K, nstart = 1)$tot.withinss
})
```

```
## [1]  85.00405  85.00405 285.40263  85.00405  85.00405
```

Diagnostic. Run `kmeans` with the same data and varying seeds; differing `tot.withinss` signals different local minima.

Fix. `kmeans(X, K, nstart = 25)`; or K-means++ via `ClusterR::KMeans_rcpp(initializer = "kmeans++")`.

7.4 Pitfall 4: K-means assumes spherical equal-variance clusters

K-means partitions space by Voronoi cells around centroids, equivalent to spherical-equal-variance Gaussian-mixture EM (§3.3). Non-spherical clusters are split or merged badly.

Tempting because: K-means is the default clustering tool, applied without checking cluster geometry.

```
set.seed(7); n <- 200
```

```r
theta_a <- runif(n, 0, pi); theta_b <- runif(n, pi, 2 * pi)
X1 <- cbind(cos(theta_a),       sin(theta_a))       + matrix(rnorm(2 * n, sd = 0.1), n, 2)
X2 <- cbind(0.5 + cos(theta_b), 0.5 + sin(theta_b)) + matrix(rnorm(2 * n, sd = 0.1), n, 2)
X <- rbind(X1, X2); true <- rep(1:2, each = n)
km <- kmeans(X, 2, nstart = 25)$cluster
table(true, km)                          # K-means cuts each moon, fails to recover
```

```
##      km
## true   1   2
##    1 148  52
##    2  54 146
```

Diagnostic. Visualize data alongside cluster boundaries; if true clusters are visibly non-spherical, K-means is wrong.

Fix. Spectral clustering via `kernlab::specc` (Ng et al., 2002); for variable-density clusters, HDB-SCAN via `dbscan::hdbscan`.

7.5 Pitfall 5: Subjectivity of the elbow method

The visual "elbow" in a scree plot or in within-cluster-sum-of-squares-vs-K is observer-dependent. On data without strong cluster structure, no clear elbow exists, but observers still report one.

Tempting because: visual inspection feels objective; the elbow heuristic is taught everywhere.

```r
set.seed(7); n <- 200
X <- matrix(rnorm(n * 5), n, 5)              # pure noise: no true cluster structure
wss <- sapply(1:10, function(k) kmeans(X, k, nstart = 25)$tot.withinss)
round(wss, 1)                                 # decrement is gradual, no clear elbow
```

```
##  [1] 958.8 805.6 696.7 623.4 563.5 513.7 475.6 444.7 421.0 400.2
```

```r
round(diff(wss), 1)
```

```
## [1] -153.2 -108.9  -73.3  -59.9  -49.8  -38.1  -30.9  -23.7  -20.9
```

Diagnostic. The right tool depends on which "k" is being chosen. For **PCA** (number of components), compute cumulative variance, Kaiser, and parallel analysis. For **K-means** (number of clusters), compute the gap statistic and silhouette. The two families are not interchangeable: parallel analysis tests a PCA eigenvalue against a random-data baseline; the gap statistic tests a clustering objective against a random-data baseline.

Fix. For PCA, parallel analysis via `paran::paran` or a pre-specified cumulative-variance threshold. For K-means, the gap statistic via `cluster::clusGap` (Tibshirani et al., 2001).

7.6 Pitfall 6: Linkage and metric sensitivity in hierarchical clustering

Different linkages produce qualitatively different dendrograms on the same data. The same dataset can support a 2-cluster, 4-cluster, or 7-cluster reading depending on linkage.

Tempting because: `hclust(d, method = "complete")` is the R default; users accept it without trying alternatives.

```r
set.seed(7); n <- 50
X <- rbind(matrix(rnorm(n, mean = 0, sd = 1.5), n / 2, 2),
           matrix(rnorm(n, mean = 2, sd = 1.5), n / 2, 2))   # overlapping clusters
d <- dist(X)
sapply(c("single", "complete", "average", "ward.D2"), function(m)
  table(cutree(hclust(d, method = m), k = 2)))    # cluster sizes vary by linkage
```

```
##   single complete average ward.D2
## 1     49       40      48      25
## 2      1       10       2      25
```

Diagnostic. Apply multiple linkages and tabulate cluster sizes or co-cluster matrices; substantial disagreement means the result is linkage-driven.

Fix. Pick linkage from prior knowledge: Ward for compact balanced clusters, single for chain-like, complete for ball-shaped. Or compute multiple and report disagreement.

7.7 Pitfall 7: Euclidean distance on categorical features

One-hot encoding categorical features and applying Euclidean distance produces nonsensical pairwise distances: any two distinct categories are at the same distance, regardless of conceptual proximity.

Tempting because: `model.matrix` encoding is one line, and the result feels numerical.

```r
library(cluster); library(clustMixType)
df <- data.frame(
  height = c(170, 165, 180, 175, 162, 178),
  region = factor(c("North", "South", "North", "East", "South", "East"))
)
df_dummy <- model.matrix(~ . - 1, data = df)
dist(df_dummy)                                    # Euclidean on dummies: meaningless
```

```
##           1         2         3         4         5
## 2  5.196152
## 3 10.000000 15.066519
## 4  5.196152 10.099505  5.196152
## 5  8.124038  3.000000 18.055470 13.076697
## 6  8.124038 13.076697  2.449490  3.000000 16.062378
```

```r
daisy(df, metric = "gower")                          # Gower handles mixed types properly
```

```
## Dissimilarities :
##            1          2          3          4          5
## 2 0.63888889
## 3 0.27777778 0.91666667
## 4 0.63888889 0.77777778 0.63888889
## 5 0.72222222 0.08333333 1.00000000 0.86111111
## 6 0.72222222 0.86111111 0.55555556 0.08333333 0.94444444
##
## Metric :  mixed ;  Types = I, N
## Number of objects : 6
```

```r
set.seed(7)
kproto(df, k = 2, verbose = FALSE)$cluster  # k-prototypes partition on mixed types
```

```
## 1 2 3 4 5 6
## 2 1 2 2 1 2
```

Diagnostic. If any feature is categorical (`class(x)` returns `factor` or `character`), Euclidean distance on a one-hot encoding is wrong.

Fix. Gower distance via `cluster::daisy(x, metric = "gower")` (Gower, 1971); or k-prototypes (the mixed-type generalization of k-means) via `clustMixType::kproto`, illustrated above.

7.8 Pitfall 8: t-SNE distances and cluster sizes

t-SNE preserves local neighbor structure; it does **not** preserve inter-cluster distances or cluster sizes. The resulting plot's between-cluster spacing and cluster diameters reflect optimization artifacts more than data structure.

Tempting because: the plot looks like a 2D projection, inviting distance interpretations as in PCA.

```
library(Rtsne)
set.seed(7); n <- 100
centers <- matrix(c(0, 0,  10, 0,  100, 0), ncol = 2, byrow = TRUE)
true <- sample(3, n * 3, replace = TRUE)
X <- centers[true, ] + matrix(rnorm(n * 3 * 2, sd = 1), n * 3, 2)
em <- Rtsne(X, perplexity = 30)$Y
round(sapply(split.data.frame(em, true), colMeans), 2)   # embedding ignores 10/100 ratio
```

```
##            1      2     3
## [1,] -14.89  -4.46 18.19
```

```
## [2,]  11.43 -14.93  5.67
```

Diagnostic. Compute true (original-space) inter-cluster distances and embedding inter-cluster distances; if their ratios disagree by a large factor, the plot's distances are not informative.

Fix. Treat t-SNE as a connectivity visualizer, not a metric-preserving projection. For distance-based downstream analysis, use PCA.

7.9 Pitfall 9: t-SNE perplexity at extremes

Very small perplexity (< 5) fragments true clusters into micro-clusters and produces near-noise embeddings; very large perplexity blurs fine local detail. The recommended range is $[5, 50]$; `Rtsne` additionally caps perplexity at $(n-1)/3$ to keep the binary search well-posed.

Tempting because: perplexity looks like a regularization knob; users sweep without understanding the failure modes.

```r
library(Rtsne); library(cluster)
set.seed(7); n <- 100
centers <- matrix(c(0, 0,  4, 0,  0, 4), ncol = 2, byrow = TRUE)
true <- sample(3, n * 3, replace = TRUE)
X <- centers[true, ] + matrix(rnorm(n * 3 * 2, sd = 0.5), n * 3, 2)
```

```r
em_low <- Rtsne(X, perplexity = 2 )$Y    # fragments true clusters
em_ok  <- Rtsne(X, perplexity = 30)$Y    # recovers them
em_hi  <- Rtsne(X, perplexity = 90)$Y    # near-uniform blob
# Silhouette of true labels on each embedding: larger = clearer separation
sapply(list(low = em_low, ok = em_ok, hi = em_hi),
       function(em) round(mean(silhouette(true, dist(em))[, 3]), 3))
```

```
##   low    ok    hi
## 0.053 0.827 0.934
```

Diagnostic. Run with multiple perplexities (e.g., 5, 30, 50); if cluster structure varies dramatically, the result is perplexity-driven.

Fix. Stay in $[5, 50]$, the standard recommended range; report multiple perplexity runs side by side.

7.10 Pitfall 10: UMAP global structure is approximate

UMAP preserves more global structure than t-SNE, but `n_neighbors` and `min_dist` parameters reshape the embedding's global geometry.

Tempting because: default parameters produce visually pleasing embeddings, suggesting they are reliable.

```r
library(uwot); library(cluster)
set.seed(7); n <- 100
centers <- matrix(c(0, 0,  4, 0,  0, 4), ncol = 2, byrow = TRUE)
true <- sample(3, n * 3, replace = TRUE)
X <- centers[true, ] + matrix(rnorm(n * 3 * 2, sd = 0.5), n * 3, 2)
em_local  <- umap(X, n_neighbors = 3,   min_dist = 0.001)  # very local
em_global <- umap(X, n_neighbors = 100, min_dist = 0.5)    # very global
sapply(list(local = em_local, global = em_global),
       function(em) c(spread = diff(range(em)),
                      sil = round(mean(silhouette(true, dist(em))[, 3]), 3)))
```

```
##            local   global
## spread 35.61316 25.55577
## sil     0.44200  0.85700
```

Diagnostic. Sweep **n_neighbors** over (e.g.) 5, 15, 50; if cluster relationships vary, global structure is parameter-driven.

Fix. Use UMAP for visualization with documented parameters; for global-structure claims, complement with PCA and quantitative validation.

7.11 Pitfall 11: Cluster validation without ground truth

Without labels, "best k" cannot be measured directly. Intrinsic measures (silhouette, gap statistic, BIC) provide indirect signals that often disagree.

Tempting because: a silhouette score is fast and produces a single number.

```r
library(cluster)
set.seed(7); n <- 200
X <- matrix(rnorm(n * 3), n, 3)
sil <- sapply(2:8, function(k) {
  km <- kmeans(X, k, nstart = 25)
  mean(silhouette(km$cluster, dist(X))[, 3])
})
gap <- clusGap(X, kmeans, K.max = 8, B = 50, nstart = 25, verbose = FALSE)
which.max(sil) + 1                                  # silhouette pick
```

```
## [1] 4
```

```r
maxSE(gap$Tab[, 3], gap$Tab[, 4])                   # gap-statistic pick
```

```
## [1] 1
```

Diagnostic. Compute multiple cluster-validation metrics; if silhouette (Rousseeuw, 1987) and the gap statistic (Tibshirani et al., 2001) disagree, the data may not have stable cluster structure.

Fix. Report multiple metrics with their disagreement explicitly. If a labeled subset exists, compute Adjusted Rand Index via `mclust::adjustedRandIndex` as external validation.

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